



FOL- CWDM

COARSE WAVELENGTH DIVISION MULTIPLEXING

EXPERIMENTAL MANUAL

.... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. FIBER OPTIC COMMUNICATION is one of the important subject need to teach while learning electronics. It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, We are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have a complete, truly innovative & superior training products we are so committed to quality that we back our products with a complete comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

Sr.No.15/8/1,Unit No. 9 Kruti Industrial Estate,
Opp Karishma Society, Off Karve Road,Kothrud,
Pune – 411038,India

ContactNo.: +91-7447438443

Email: lab@akademika.in

www.akademika.in

SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
- Before connecting the power supply to the kit, be sure that connecting chords are connected correctly, as per the experiment.
- In Advance Fiber Optic Experimental Modules, signals are to be monitored with a Digital Storage Oscilloscope, or with the help of indicators as explained in the manual. The scope input channel should be D.C. coupled, unless otherwise indicated. For observation of signals on oscilloscope use X10 (Attenuation Probe).

WARRANTY

These kits are warranted against defects in workmanship and materials. If any failure resulting from a defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting patch chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the event warranty service is needed, purchaser should get in touch with service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit /system has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs and maintenance of old and decrepit products, be sure to contact our service center or sales outlet.

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TECHNICAL SPECIFICATION

LASERS

- 1.25 Gbps CWDM Laser Diode Modules at wavelengths of 1510nm, 1530nm, 1550nm, 1570nm
- In built Isolator
- Channel spacing : 20 nm
- Threshold Current I_{th} : 10 mA Typical
- Output power : @ $I_{th} + 30$ mA - > 0.7 mW
@ ~ 58 mA - > 1.4 mW
- Operating Voltage : 1.1V Typical

DETECTORS

- 1.5 GHz InGaAs PIN Photodiode Module.
- Responsively : Typical 0.9 A/W in 9/125 μ m fiber
- Spectral Range : 1250nm to 1600nm
- Reverse Voltage : 30 V max.

CWDM multiplexer and de-multiplexer (4 channels)

- Center Wavelength : 1510nm, 1530nm, 1550nm & 1570nm
- Channel spacing : 20nm
- Pass band @ 0.5dB : ITU+/- 6.5 nm
- Insertion Loss @ MUX / DEMUX Port : <= 2.9 dB
- Adjacent Channel Isolation : >= 30 dB
- Non Adjacent Channel Isolation : >= 40 dB
- Max Optical Power : 300 mW

Three Port Circulator

- Polarization Independent Optical Circulator
- Ports : 3 Nos.
- Band : C+L
- Wavelength Range : 1525nm to 1610nm
- Transmission Direction : 1 -> 2 , 2 -> 3
- Channel Isolation : > 40 dB

- Insertion loss : ≤ 0.9 dB

Fiber Bragg grating

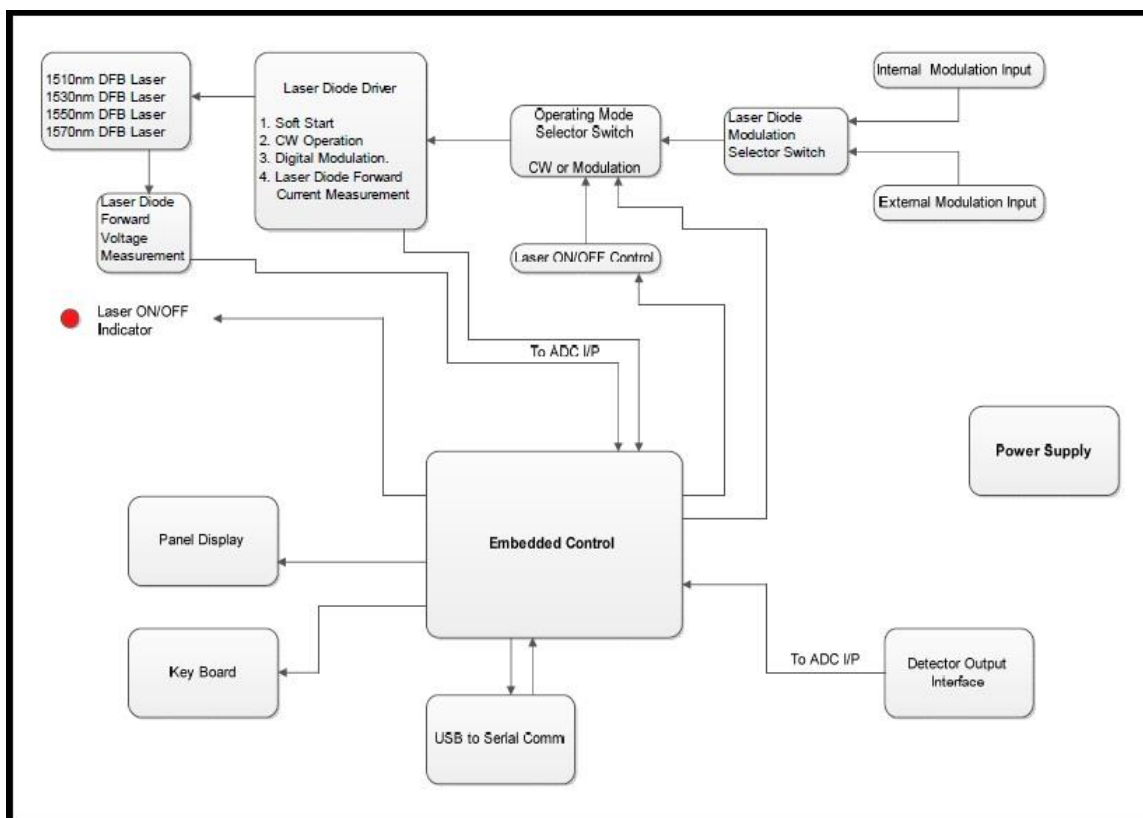
- Central Wavelength : 1553 ± 0.5 nm
- Bandwidth @ 3 dB : 0.02 – 5 nm
- SLSR : > 15 dB
- Reflectivity : $> 90\%$

Software

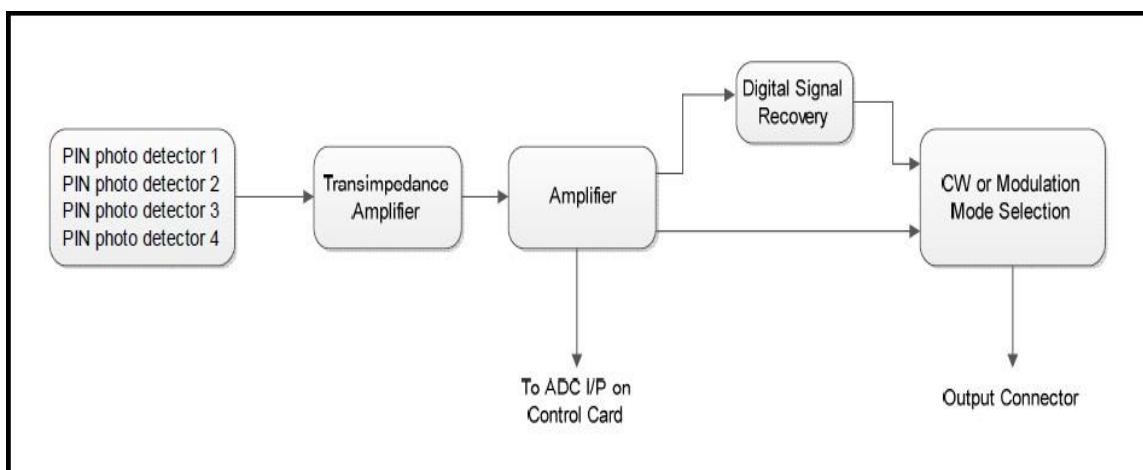
- User friendly GUI for monitoring, controlling of CWDM system
- Operating modes like CW mode, VI characteristics mode, Internal & External Modulation.
- LASER controls like Supply ON/OFF, wavelength selection & driving current selection.
- Real time output signal monitoring of Photo-detector.
- Graphical representation : XY plot of VI characteristics & Internal Modulation.
- COM Settings : USB 2.0
- Operating System : XP, VISTA, Windows7, Windows8.

Functional Block Diagram

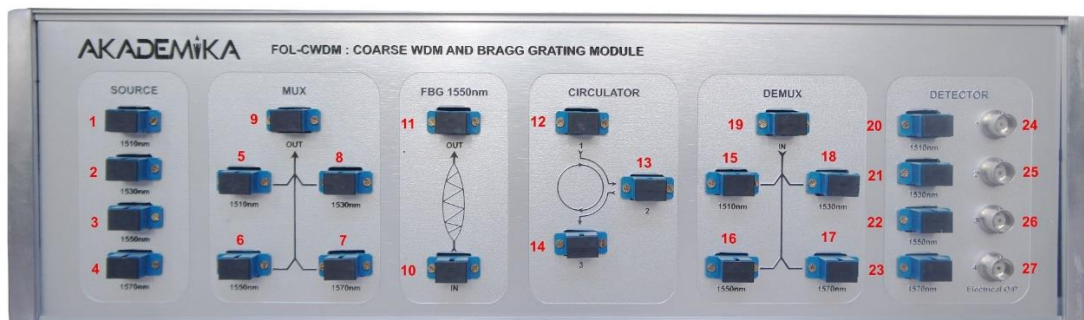
Transmitter



Receiver



FRONT PANEL DETAILS



- | | |
|----|---------------------------------------|
| 1 | LASER Source (Optical OUT) (1510 nm) |
| 2 | LASER Source (Optical OUT) (1530 nm) |
| 3 | LASER Source (Optical OUT) (1550 nm) |
| 4 | LASER Source (Optical OUT) (1570 nm) |
| 5 | MUX (Optical IN) (1510 nm) |
| 6 | MUX (Optical IN) (1550 nm) |
| 7 | MUX (Optical IN) (1570 nm) |
| 8 | MUX (Optical IN) (1530 nm) |
| 9 | MUX (Optical OUT) |
| 10 | Bragg Grating (1550 nm) (Optical IN) |
| 11 | Bragg Grating (1550 nm) (Optical OUT) |
| 12 | Circulator Terminal-1 |
| 13 | Circulator Terminal-2 |
| 14 | Circulator Terminal-3 |
| 15 | DeMUX (Optical OUT) (1510 nm) |
| 16 | DeMUX (Optical OUT) (1550 nm) |

17	DeMUX (Optical OUT) (1570 nm)
18	DeMUX (Optical OUT) (1530 nm)
19	DeMUX (Optical IN)
20	Detector 1 (Optical IN)
21	Detector 2 (Optical IN)
22	Detector 3 (Optical IN)
23	Detector 4 (Optical IN)
24	Detector 1 (Electrical O/P)
25	Detector 2 (Electrical O/P)
26	Detector 3 (Electrical O/P)
27	Detector 4 (Electrical O/P)

BACK PANEL DETAILS



- | | |
|---|----------------------------------|
| 1 | External I/O 1 |
| 2 | External I/O 2 |
| 3 | External I/O 3 |
| 4 | External I/O 4 |
| 5 | USB |
| 6 | Power Supply Switch and AC Input |

DELIVERABLES

○ FOL-CWDM System	-	01
○ Shielded USB A-B cable	-	01
○ Power Cable	-	01
○ SC/PC –SC/PC Single Mode Fiber Optic Patch Chords	-	12
○ BNC to BNC coaxial cable	-	04
○ FOL-CWDM Software CD	-	01
○ FTDI Drivers included on CD		

PRECAUTIONS OF USE

The kit contains four laser diodes which can emit up to 1.5 mW of power at 1.55 μm . Hence it is necessary to follow basic security instructions.

- Always use protective goggles of suitable wavelength.
- Do not switch the lasers without making it sure that all patch cords are properly plugged in.
- Do not look in to the laser beam or do not directly look in to the fiber while laser is ON.
- Do not plug faulty patch cords in to the kit.
- Do not plug or unplug the patch cords while laser is functioning.
- Clean the tips of the p
- atch cords with dry optical paper or with ACETONE only.
- Never connect two lasers together.
- Never do forceful operations

LASER SAFETY

It is imperative that you read and understand the following safety information before attempting any of the experiments from this manual. If you have inquiries, please consult your lecturer or demonstrator.

- **Operational Hazard-Semiconductor Laser Diode**

The laser diode source emits radiation at 1550nm, which is Invisible to the human eye and may cause **eye damage** if the output beam is viewed directly. While using, never stare at the optical output port of the Laser Diode, when its protective cap is removed. Always leave the protective cap for the fiber patch cord in position until ready to make a connection and never stare at the end of a fiber patch cord connected to the Laser diode output port or point the fiber end at yourself or anyone else in your vicinity before making a connection. Never use external optics (i.e. Collimating lenses) to view the Laser Diode emission at a close range.

The output power of the semiconductor Laser Diode is less than 2mW which is in line with all the applicable standards of DHMS regulations and falls within the international safety Class3B.

- This kit must be employed only for the use for which it has been conceived, i.e. as an educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.
- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.
- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

ABOUT FIBER OPTICS

Fiber optics technology is rapidly becoming a familiar & indispensable part of human life. You have undoubtedly heard of Fiber Optics. All giant telephone companies like AT & T, sprints have saturated the airways & print media with advertisements heralding this & new technology. Futurists, science writers & entrepreneurs forecast fantastic growth for Fiber Optics applications and their enthusiasm is valid. Today medical fiber optics systems allow physicians to peer inside the human body without surgery. Military commanders demand portable battlefield communication systems, which use superior fiber optic transmissions. The massive bundles of copper wiring which once carried telephone conversations between continents now are being replaced with the optical fibers from coast to coast.

Very few technologists ever come close to the fantastic growth rate that market analyst love for them to predict. Fiber optics has exceeded those rates. The next decade promises even more wonders. The influence of fiber optics will pervade our homes workspaces & recreational facilities. Your television reception will take on startling clarity. Newsprint may largely become a thing of the past as environmentally friendly electronics newspapers become feasible. Two-way fiber optics will allow you to “attend college” in the comfort of your home.

Anyone who takes an interest in fiber optics & pursues a career in the field today is on the ground floor of opportunity. The technology is sufficiently new that few experts exist. Many science colleges, universities & engineering colleges have included fiber optics in their syllabus. The future beckons existing with the prospect of making new discoveries finding new application for the technology.

This kit is an introduction to fiber optics communication technology for instructors, students & hobbyist. Welcome to fascinating & expanding world of fiber optics. We hope that you find the kit challenging, stimulating & -yes- even fun.

Before fiber optics came along the primary means of real time, data communication was electrical in nature. It was accomplished using copper wire or by transmitting electromagnetic waves through free space. Fiber optics changed that by providing an alternative means of sending information over significant distances over light energy. Although, Fiber Optics was a very controversial technology initially, today, it is considered as reliable & and cost effective.

High bandwidth properties of fiber cables can be used over greater distances because of the Low Loss as compared to copper cables. In data networks, this can be as much as 2Km without the use of repeaters. Their light weight and small size also make them ideal for applications, where running copper cables would be impractical. By using multiplexers, one fiber can replace hundreds of copper cables. This is pretty impressive for a tiny glass filament, but the real benefits in the data industry are its immunity to Electro Magnetic Interference (EMI), and the fact that glass is not an electrical conductor. Fiber is non- conductive, therefore, it can be used wherever electrical isolation is required. For instance, it can be used between buildings where copper cables would require cross bonding to eliminate differences in earth potentials. Also, Fibers do not pose any threat in dangerous environments such as

chemical plants, where a spark could trigger an explosion. Last but not least is the security aspect. It is very, very difficult to tap into a fiber cable to read the data signals.

You will learn soon that fiber optic communication is indeed a simple technology, closely related to electronics. In fact, it was research in electronics that set the stage for fiber optics to develop into the communication giant that it is today.

What is fiber-optic communication?

A fiber-optic communication system is a communication system in which the transmitter and receiver are linked by fiber-optic cable. At the transmitter, light is shone into the cable. This travels down the fiber-optic cable, and can be detected at the receiver. If we change the brightness of the light beam at the transmitter, we can detect this change in brightness at the receiver-this allows us to send information over the fiber-optic link.

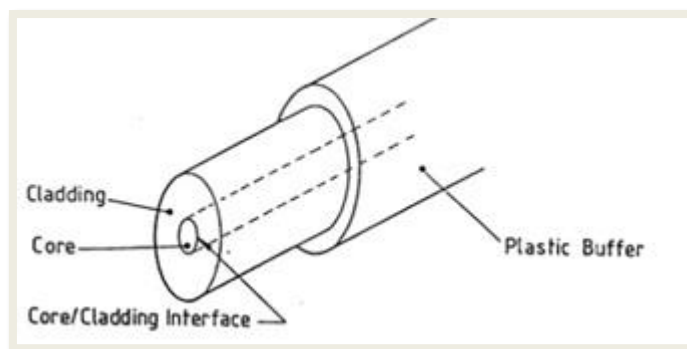
Why use light?

Fiber-optic cables have a number of advantages over metal cables when it comes to communication systems. They:

- Have smaller diameters
- Are lighter in weight and more flexible
- Allow much greater distance to be covered without any amplifiers / electronics
- Have extremely large bandwidths
- Provide electrical isolation between transmitter and receiver
- Naturally insulate between any electromagnetic wave energy i.e. it is extremely difficult to eavesdrop)
- Are immune to interference due to other people's electrical equipment.

What is an optical fiber?

An optical fiber is basically a light wave-guide, which transmits energy at optical frequencies. It consists of a central transparent core down which the light travels, surrounded by a layer of transparent cladding, which creates an optical interface at the core/cladding boundary as shown in Figure.

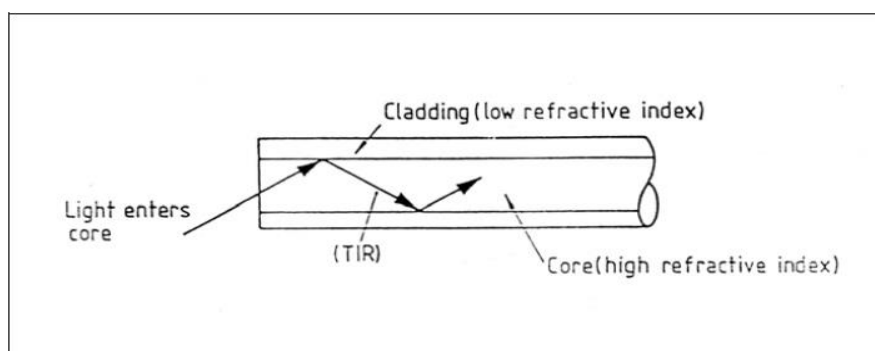


Construction of a fiber Optic cable

The core and cladding may be made from either **glass** or **plastic**, and non-transparent **plastic buffer** to provide environmental protection surrounds the cladding.

How does light travel down an optical fiber?

Light is transported through the fiber-optic core by continuously being reflected off the Core / cladding interface. This is called **total internal reflection (TIR)**, and is shown in Figure below:

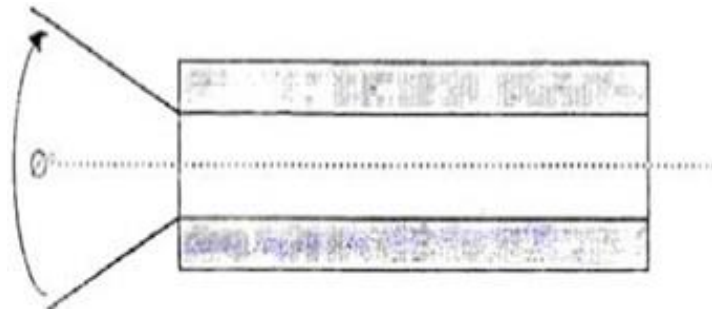


Passage of light along a fiber by total internal reflection

The reflections are made possible by having a high reflective index for the core material and a lower refractive index for the cladding material. To create TIR the core must have a higher refractive index than the cladding, and as we shall see later, the light must not strike the core/cladding interface at too steep an angle.

Acceptance angle and numerical aperture

In order for total internal reflection to take place inside an optic fiber, light rays entering the end of the fiber must fall inside an imaginary 'cone' of angle Φ° , as shown below



Although light rays outside this cone may still enter the fiber, they will hit the core-cladding interface at too steep an angle for TIR to take place, and their energy will be lost into the cladding.

At the far end of the fiber, light will emerge in a similar cone, also of angle Φ . The angle Φ is an important parameter of any fiber-optic cable, and is called the **acceptance angle** of the fiber.

To avoid conversion over units (degrees, radians, etc.), the term numerical aperture is often used in data sheets for optic fibers. The numerical aperture (NA) of a fiber is defined as the sine of half the acceptance angle, i.e. $NA = \sin(\Phi/2)$

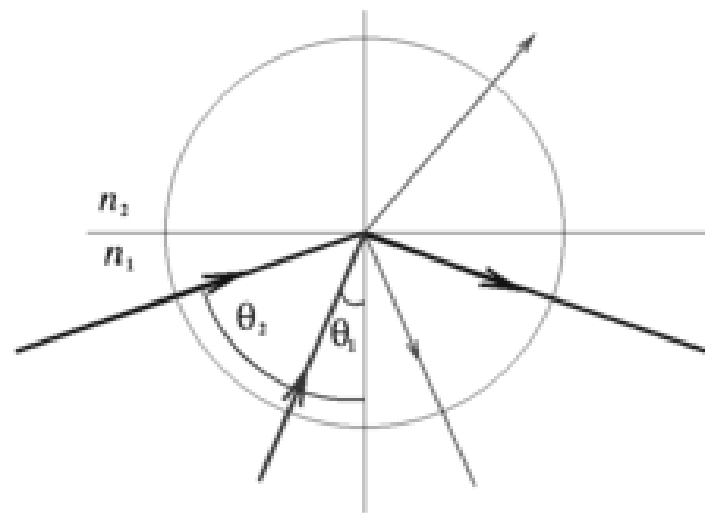
How Optical Fiber Works?

The principle of operation of optical fiber lies in the behavior of light. Light travels in straight line through most optical materials, but that's not necessarily the case at the junction (interface) of two materials of different refractive indices. Think of a fiber cable in terms of a very long cardboard roll (from the inside roll of paper towel) that is coated with a mirror. If you shine a flashlight in one you can see light at the far end - even if you bend the roll around a corner.

Light pulses move easily down the fiber-optic line because of a principle known as the total internal reflection. "This principle of the total internal reflection states that when the angle of incidence exceeds a critical value, light cannot get out of the glass; instead, the light bounces back in as shown in figure. When this principle is applied to the construction of the fiber-optic strand, it is possible to transmit information through fiber lines in the form of light pulses.

Total internal reflection is an optical phenomenon that occurs when light strikes a medium boundary at a steep angle. If the refractive index is lesser on the other side of the boundary, no light can pass through, so all of the light gets reflected effectively. The critical angle is the angle of incidence above which the total internal reflection occurs.

When light crosses a boundary between materials with different refractive indices, the light beam will be partially be refracted at the boundary surface and partially it will get reflected. However, if the angle of incidence is shallower (closer to the boundary) than the critical angle (the angle of incidence where light is refracted so that it travels along the boundary), then the light will stop crossing the boundary altogether, instead it will reflect back totally internally. This can only occur where light travels from a medium with a higher refractive index to the one with a lower refractive index. For example, it will occur when light passes from glass to air, but not when light would pass from air to glass.

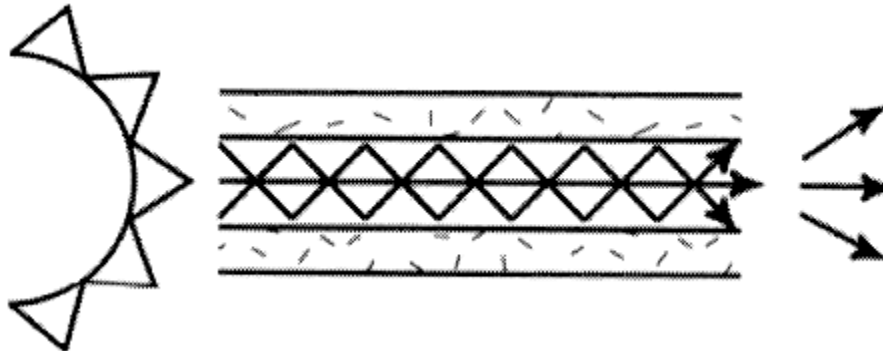


Total internal reflection

Types of Optical Fibers

The three types of fiber optic cables commonly used are as follows:

- Single Mode
- Multimode
- Plastic Optical Fiber (POF)



Fiber optic cable functions as a 'light guide', guiding the light introduced at one end of the cable to the other end. The light source can either be a light-emitting diode (LED) or a laser.

The light source is pulsed on and off, and a light-sensitive receiver on the other end of the cable converts the pulses back into the digital ones and zeros, of the original signal.

Even laser light shining through a fiber optic cable is subject to loss of strength, primarily through dispersion and scattering of the light, within the cable itself. The faster the laser fluctuates, the greater the risk of dispersion. Light strengtheners, called repeaters, may be necessary to refresh the signal in certain applications.

While fiber optic cable itself has become cheaper over time - an equivalent length of copper cable costs lesser per foot, but not in capacity. Fiber optic cable connectors and the equipment required to install them are more expensive than their copper counterparts.

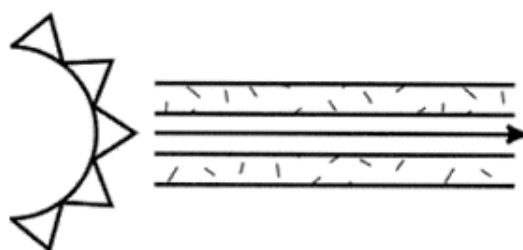
Single Mode Cable

It is a single strand of glass fiber with a diameter of 8.3 to 10 microns that has one mode of transmission. Single Mode Fiber with a relatively narrow diameter, through which only one mode will propagate typically 1310nm or 1550nm, carries higher bandwidth than multimode fiber, but requires a light source with a narrow spectral width.

Single-mode fiber gives you a higher transmission rate and up to 50 times more distance than multimode, but it also costs more. Single-mode fiber has a much smaller core than multimode. The small core and single light-wave virtually eliminate any distortion that could result from overlapping light pulses, providing the least signal attenuation and the highest transmission speeds of any fiber cable type.

Single-mode optical fiber is an optical fiber in which only the lowest order bound mode can propagate at the wavelength of interest, typically 1300nm to 1320nm.

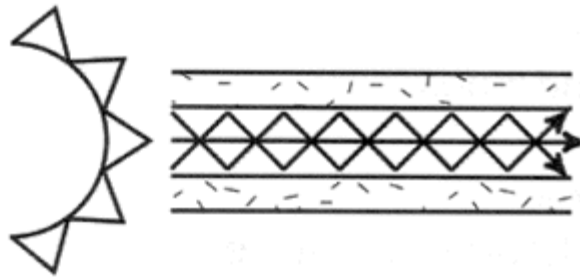
“Single mode fiber”
single path through the fiber



Multimode Cable

It is made of glass fibers, with common diameters in the 50-to-100 micron range for the light carry component (the most common size is 62.5 μ m). Multimode fiber gives you high bandwidth at high speeds over medium distances. Light waves are dispersed into numerous paths, or modes, as they travel through the cable's core, typically 850nm or 1300nm. Typical multimode fiber core diameters are 50, 62.5, and 100 micrometers. However, in long cable runs (greater than 3000 feet [914.4 meters]), multiple paths of light can cause signal distortion at the receiving end, resulting in an unclear and incomplete data transmission.

“Multimode fiber”
multiple paths through the fiber



Plastic Optical Fiber (POF)

It is a newer plastic-based cable which promises performance, similar to glass cable on very short runs, but at a lower cost.

Advantages of Fiber Optics over Conventional Copper Cables

- **Less expensive** - Several miles of optical cable can be made at cheaper rates compared to equivalent lengths of copper wires. This saves your service providers (cable TV, Internet) and your money too.
- **Thinner** - Optical fibers can be drawn to smaller diameters than copper wire.
- **Higher carrying capacity** - Optical fibers are thinner than copper wires, therefore, more fibers can be bundled within a given diameter cable than the copper wires. This allows more phone lines to go over the same cable or more channels to pass through the cable into your cable TV box.
- **Less signal degradation** - The loss of signal in optical fiber is lesser than that observed in copper wires.
- **Light signals** - Unlike electrical signals in copper wires, light signal from one fiber does not interfere with those of the other fibers in the same cable. This means clearer phone conversations or TV reception.
- **Low power** - Signals in optical fibers degrade less; therefore, lower-power transmitters can be used instead of the high-voltage electrical transmitters, as required for copper wires. Again, this saves your provider and your money.
- **Digital signals** - Optical fibers are ideally suited for carrying digital information, which is especially useful in computer networks.
- **Non-flammable** - No electricity is passed through optical fibers; therefore, there is no fire hazard.
- **Lightweight** - An optical cable weighs lesser than a comparable copper wire cable. Also, Fiber-optic cables take very little space in the ground.
- **Flexible** - Fiber optics is very flexible and can transmit and receive light effectively; therefore, they are used in many flexible digital cameras for the various purposes. Some of the examples are mentioned below:

- **Medical imaging** – Used in bronchoscopes, endoscopes and laparoscopes.
- **Mechanical imaging** – To inspect mechanical welds along pipes and engines (airplanes, rockets, space shuttles, cars).
- **Plumbing** - To inspect sewer lines.

With all the above properties of optical fiber, it is becoming very important in sensing fields too. Most of the properties of optical fiber are useful in sensors. Further, optical fiber sensors offer extreme sensitivity & can be used to sense almost any physical parameters like temperature, velocity, current, voltage etc.

INTRODUCTION CWDM

Coarse wavelength-division multiplexing (CWDM) technology was first commercially deployed in the early 1980s for transporting digital video signals over multimode fiber. Quante Corp. marketed a system in the 800 nm window with four channels running 140 Mb/sec. each. These systems were primarily used in cable TV links. However, CWDM systems did not generate significant interest among service providers - until now.

As metro carriers seek cost-effective solutions to their transport needs, CWDM is becoming more widely accepted as important transport architecture. Unlike DWDM, systems based on CWDM technology deploy **uncooled distributed-feedback (DFB) lasers** and wideband optical filters. These technologies provide several advantages to CWDM systems such as lower power dissipation, smaller size, and less cost. The commercial availability of CWDM systems offering these benefits makes the technology a viable alternative to DWDM systems for many metro and access applications.

CWDM and DWDM

The bandwidth of a fiber-optic link can be increased by transmitting data faster, or transmitting multiple wavelengths in a single fiber, with a method known as **WDM (Wavelength Division Multiplexing)**. Wavelength Division Multiplexing Technology involves using a multiplexer to combine wavelengths traveling on different fibers into a single fiber. At the receiver end of the link, a demultiplexer separates the wavelengths and routes them into different fibers, which all terminate at separate receivers (see **Figure 1.**). The space between the individual wavelengths transmitted through the same fiber serve as the basis for defining DWDM and CWDM (see **Figure 2.**).

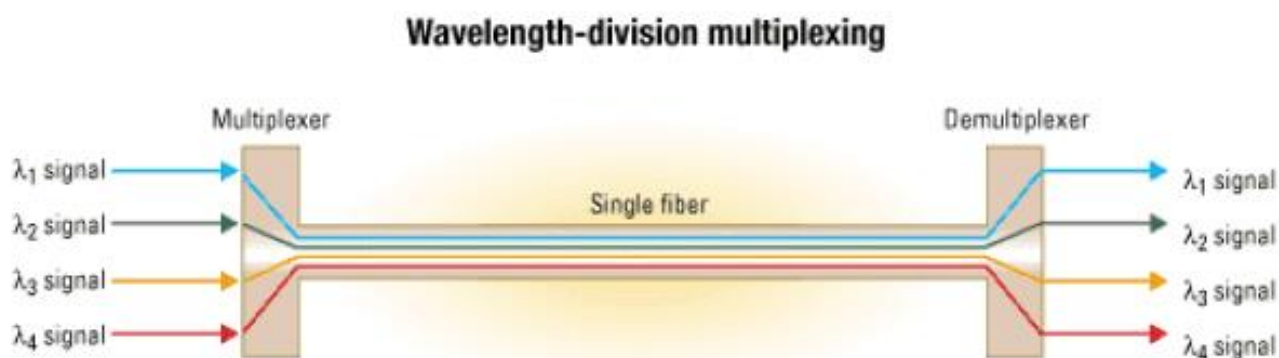


Figure 1

Figure Shows WDM is accomplished using a multiplexer to combine wavelengths traveling on separate fibers into a single fiber. At the receiver end of the link, a demultiplexer separates the wavelengths and routes them into different fibers.

DWDM systems typically use one of three possible wavelength separations:

200 GHz (1.6 nm)

100 GHz (0.8 nm)

50 GHz (0.4 nm)

Future systems are projected to have even narrower spacing. The operating wavelengths of DWDM systems are defined according to a standardized frequency/wavelength grid developed by the **International Telecommunication Union** (<http://www.itu.int>).

Figure 2 The spacing between the individual wavelengths transmitted through the same fiber serve as the basis for defining DWDM and coarse WDM (CWDM). Spacing in CWDM systems is typically 20 nm, while most DWDM systems today offer 0.8-nm (100-GHz) wavelength separation.

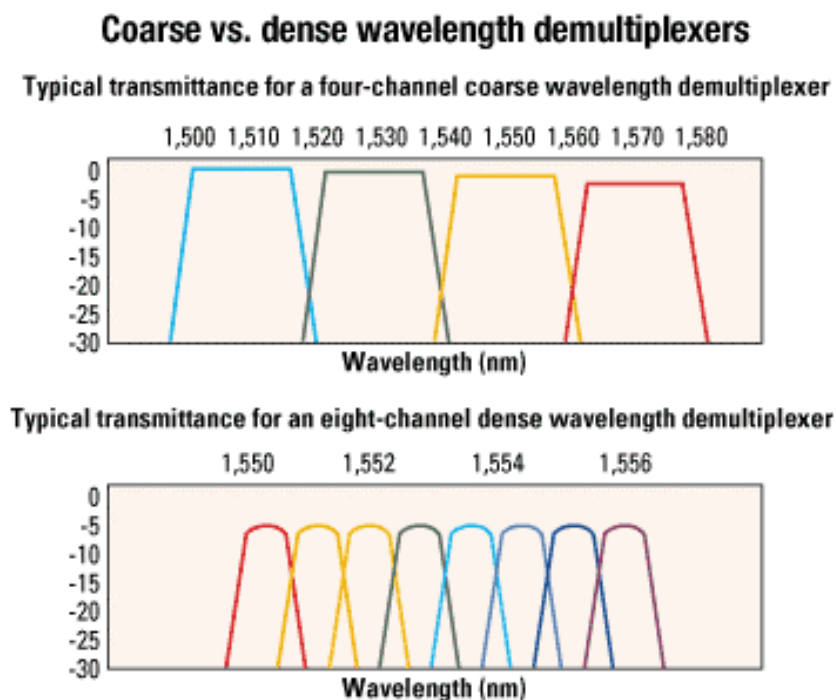


Figure 2

Both Coarse and Dense architectures utilize **Distributed Feedback Lasers (DFB)**. However, DWDM systems require the larger *cooled* DFB Lasers. This is because a semiconductor laser wavelength drifts about 0.08 nm/°C with temperature. DFB lasers are cooled to stabilize the wavelength from drifting outside the pass band of the multiplexer and de-multiplexer filters as the temperature fluctuates in DWDM systems.

CWDM systems, on the other hand, use DFB lasers that are *not* cooled. These systems typically operate from 0 to 70°C with the laser wavelength drifting about 6 nm over this range. This wavelength drift, coupled with the variation in laser wavelength of up to ± 3 nm (due to laser die manufacturing processes), yields a total wavelength variation of about ± 12 nm. The optical filter pass band and laser channel spacing must be wide enough to accommodate the

wavelength variation of the uncooled lasers in CWDM systems (see **Figure 3**). Channel spacing in these systems is typically 20 nm with channel bandwidth of 13 nm.

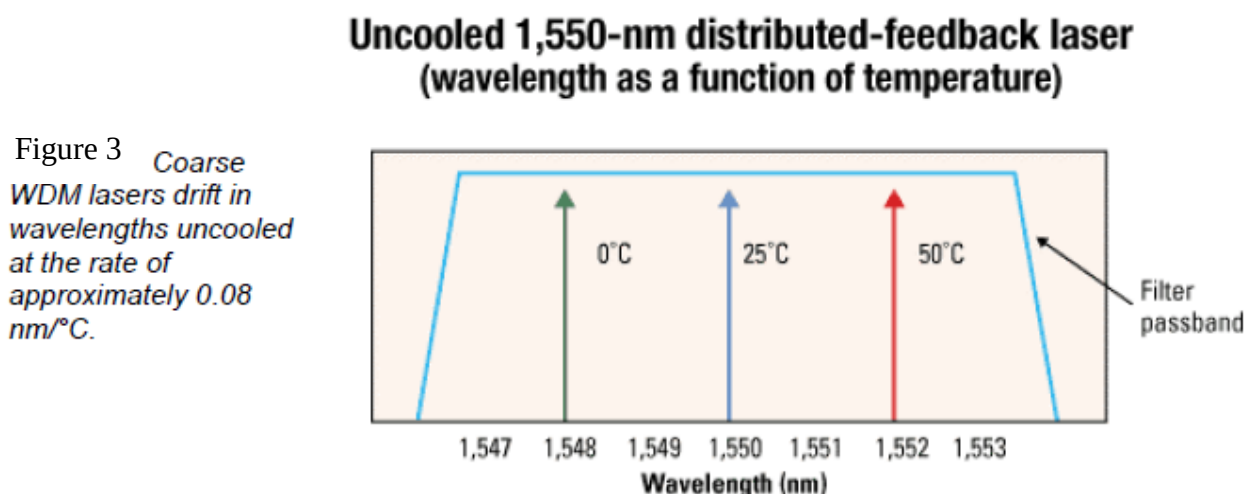


Figure 3

Coarse WDM lasers drift in wavelengths uncooled at the rate of approximately 0.08 nm/°C

Less Expensive Hardware

The cost difference between CWDM and DWDM systems can be attributed to hardware and operating costs. While DWDM lasers are more expensive than CWDM lasers, the cooled DFB lasers provide cost-effective solutions for long-haul transport and large metro rings requiring high capacity. In both of these applications, the cost of the DWDM systems is amortized over the large number of customers served by these systems. Metro-access networks, on the other hand, require lower-cost and lower-capacity systems to meet market requirements, which are based largely on what the customer is willing to pay for broadband services.

The price of DWDM transceivers is typically four or five times more expensive than that of their CWDM counterparts. The higher DWDM transceiver costs are attributed to a number of factors related to the lasers. The manufacturing wavelength tolerance of a DWDM laser die compared to a CWDM die is a key factor. Typical wavelength tolerances for DWDM lasers are on the order of ± 0.1 nm; whereas tolerances for CWDM laser die are ± 2 -3 nm. Lower die yields also drive up the costs of DWDM lasers relative to CWDM lasers. In addition, packaging DWDM laser die for temperature stabilization with a Peltier cooler and thermistor in a butterfly package is more expensive than the uncooled CWDM coaxial laser packaging.

The cost difference between DWDM and the CWDM multiplexers and de multiplexers contributes to lower overall system costs in favor of CWDM as well. CWDM filters are inherently less expensive to make than DWDM filters due to the fewer number of layers in the filter designs. Typically, there are approximately 150 layers for 100-GHz filter designs

used in DWDM systems, whereas there are approximately 50 layers in a 20-nm, CWDM filter. The result is higher manufacturing yields for CWDM filters. In the manufacture of CWDM three-port devices, the alignment tolerances are relaxed relative to those for DWDM devices so that the manufacturing costs for CWDM devices are significantly less (see **Figure 4**).

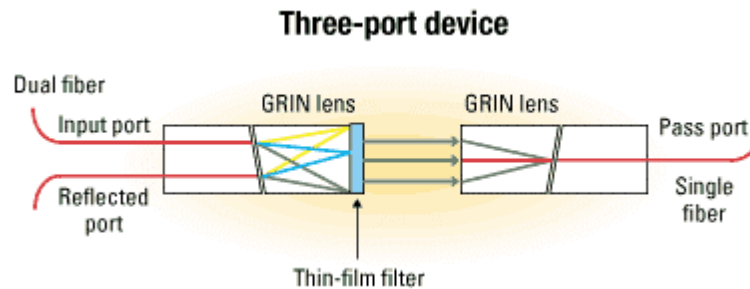


Figure 4

Figure shows Coarse WDM (CWDM) filter designs use fewer layers, and the alignment tolerances are relaxed relative to those for DWDM devices, which significantly lowers the manufacturing costs of CWDM devices.

The CWDM filter costs about 50% less than the DWDM filter and is projected to drop by a factor of three in the next two to three years as automated manufacturing takes hold. The adoption of new filter and multiplexing /demultiplexing technologies is expected to decrease costs even further.

SOFTWARE INSTALLATION

I. SYSTEM REQUIREMENTS:

- Supporting Operating System :
 - Windows XP (Service Pack 3)
 - Window 7 (32/64 bit)
 - Windows Vista (32/64 bit)
 - Windows 8 (32/64 bit)
 - Windows 8.1 (32/64 bit)
- Software Requirements :
 - **NI LabVIEW Runtime Engine 2013 SP1** (Included in setup.exe)
 - **NI - VISA Runtime 5.4** (Included in setup.exe)

- Minimum Hardware Requirements :

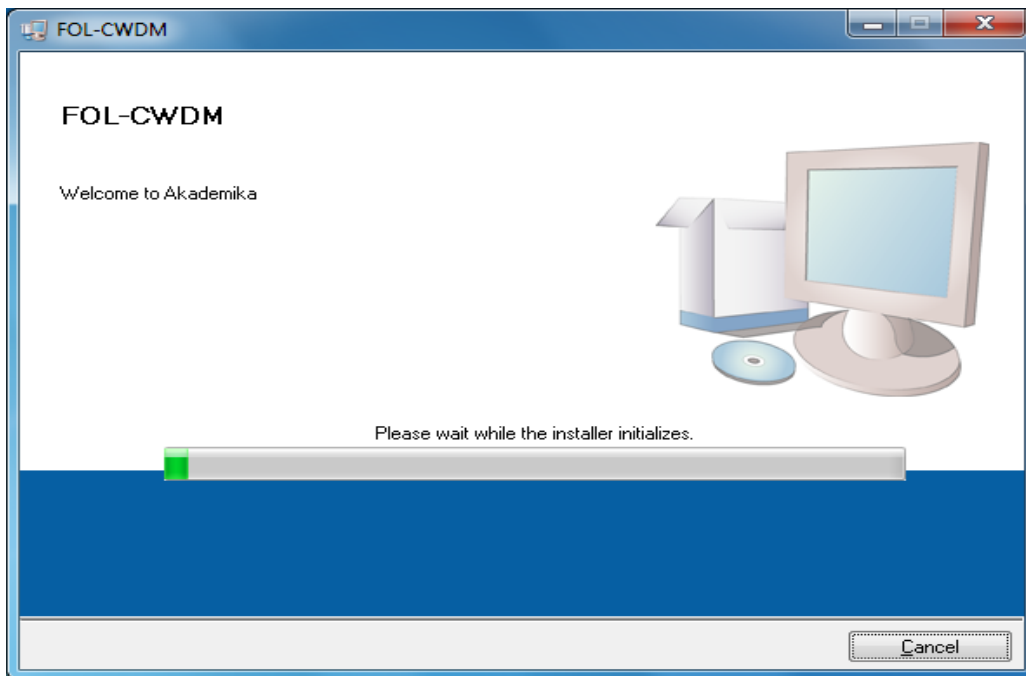
Processor	:	PENTIUM-4
Frequency	:	1 GHz
RAM	:	1 GB RAM
Disk Space	:	Minimum 300 MB free
Screen Resolution	:	1024*768
Graphics	:	256 MB

- System & Accessories list:
 - FOL-CWDM System
 - USB A-B cable
 - Power Cable
 - Fiber Optic Cables
 - BNC to BNC coaxial cable
 - FOL-CWDM Software
 - FTDI Drivers
 - Adobe Reader for windows

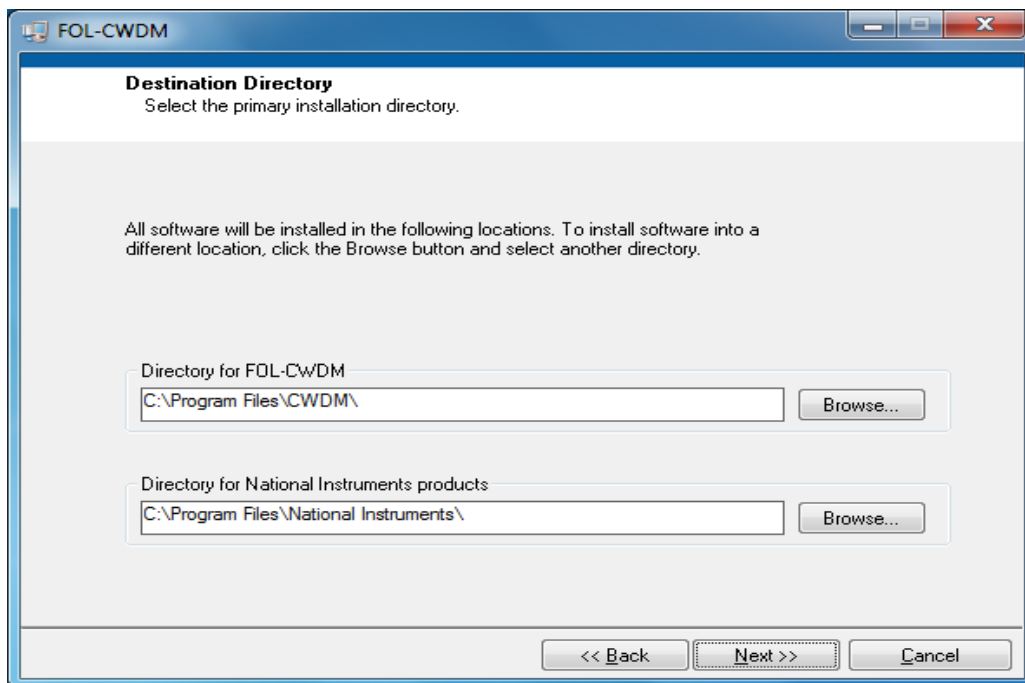
- VLC media player for Windows

II. SOFTWARE INSTALLATION PROCEDURE:

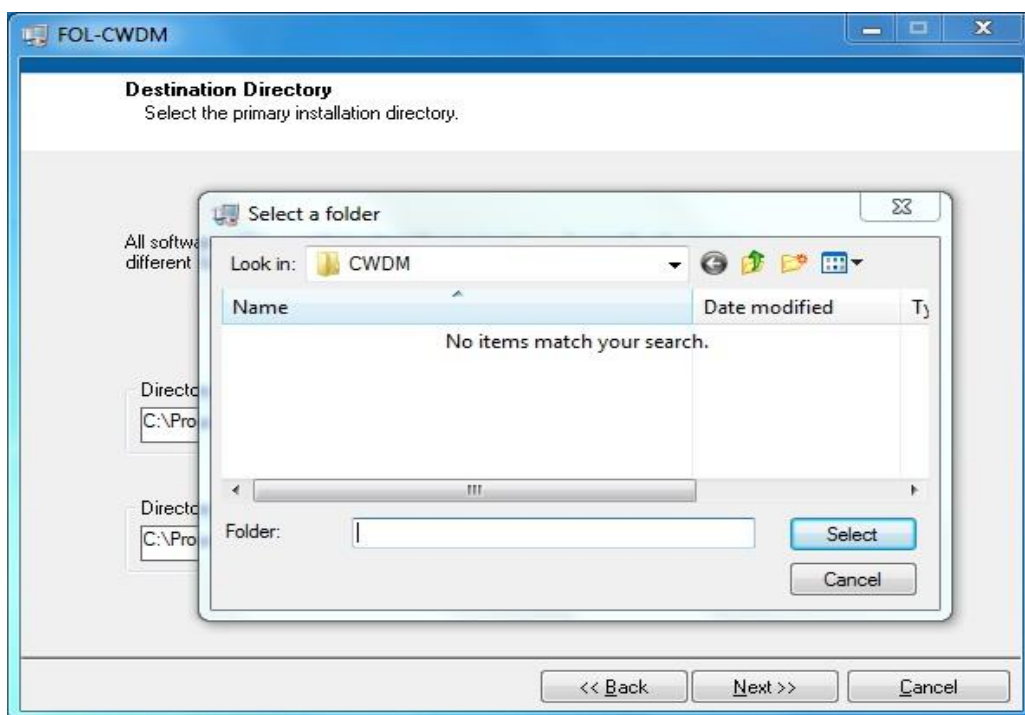
1. Locate folder “FOL-CWDM Installer” -> “Volume” provided in CD and run **Setup.exe** from this folder. Select the directory to install and click on **NEXT**. A welcome window will appear.



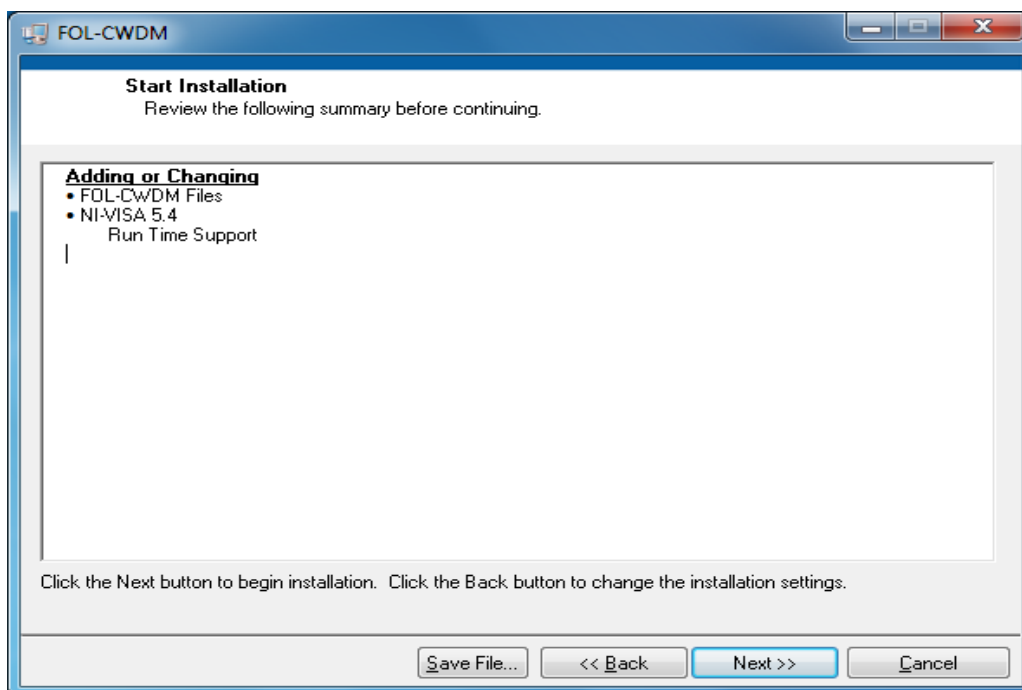
2. Select the directory to install. Default path is **C:\Program Files\CWDM**.



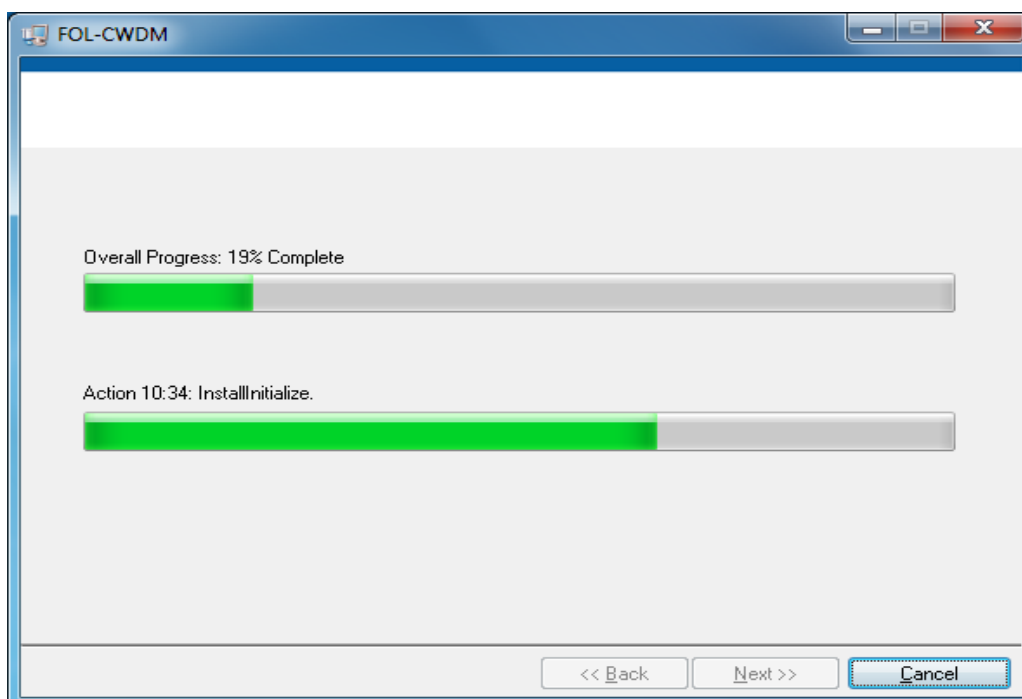
You can change the path where to install the software.



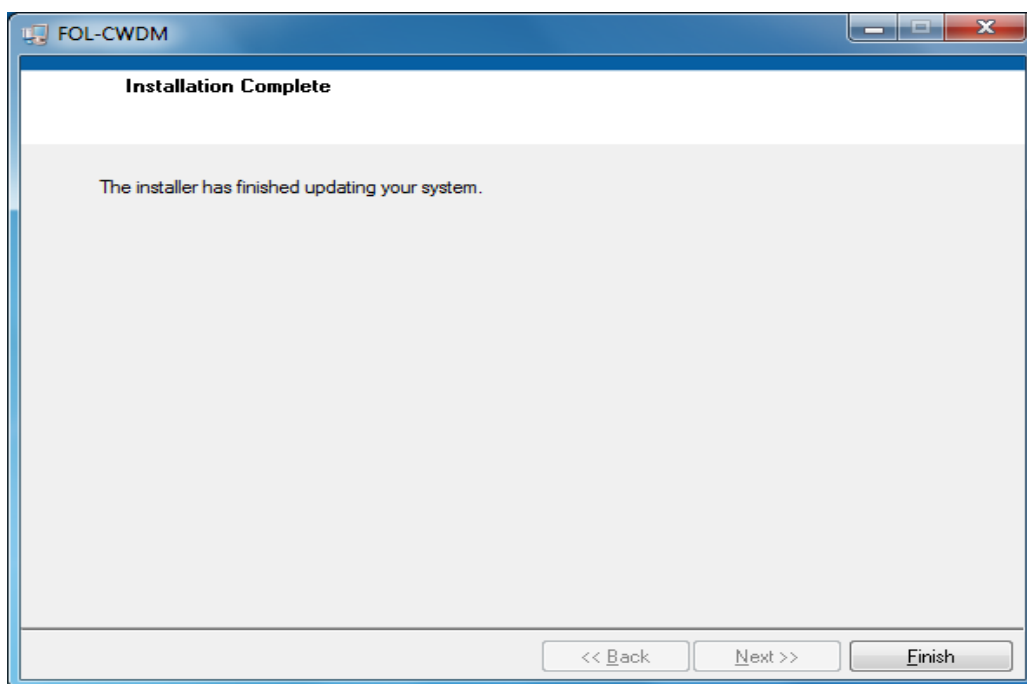
3. It will show which things will be going to install.



4. Wait for the installation to complete.



5. Finish the Setup. The Software is now ready to use.



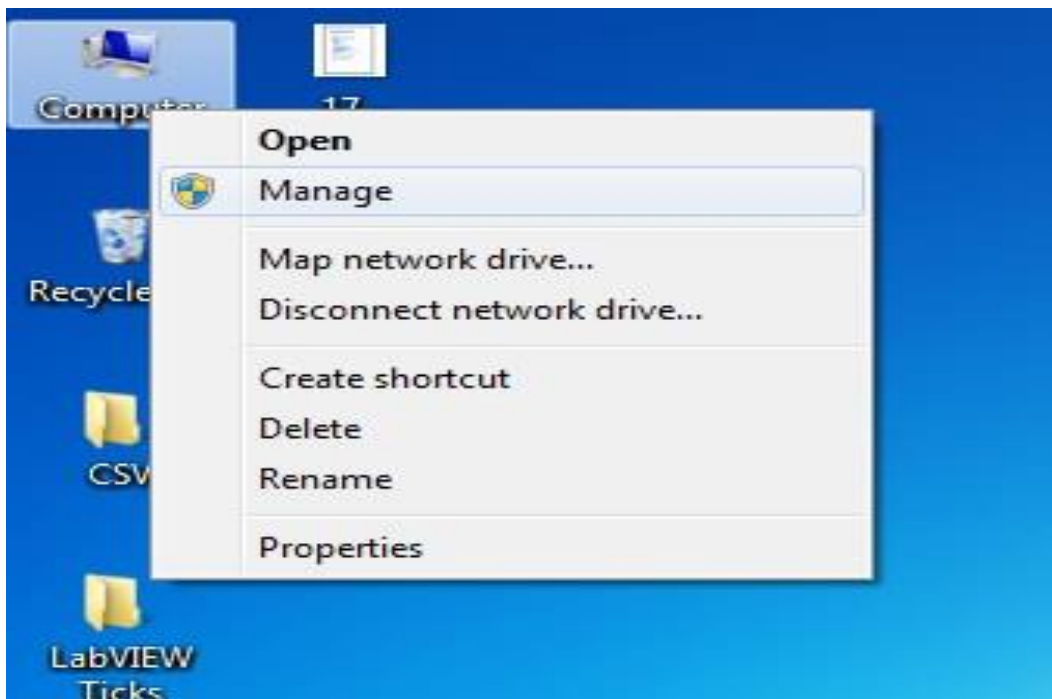
III. INSTALLING THE DRIVERS:

1. Connect the FOL-CWDM Trainer Kit to PC via USB-A to B cable after Turning ON the device.
2. To install the drivers, find a folder named '**FTDI Drivers**' provided in the CD, again four folders are inside there, according to the various Operating Systems. These folders contain drivers for USB to Serial Bridge.

Below are the folders inside FTDI Drivers;

- Windows- 8.1 (64 bit)
- Windows- 8.1 (32 bit)
- Windows- 8 (64 bit)
- Windows- 8 (32 bit)
- Windows-7 (32 bit)
- Windows-7 (64 bit)
- Windows XP (64 bit)
- Windows XP (32 bit)

Right click on **My Computer** Icon and select **Manage**.

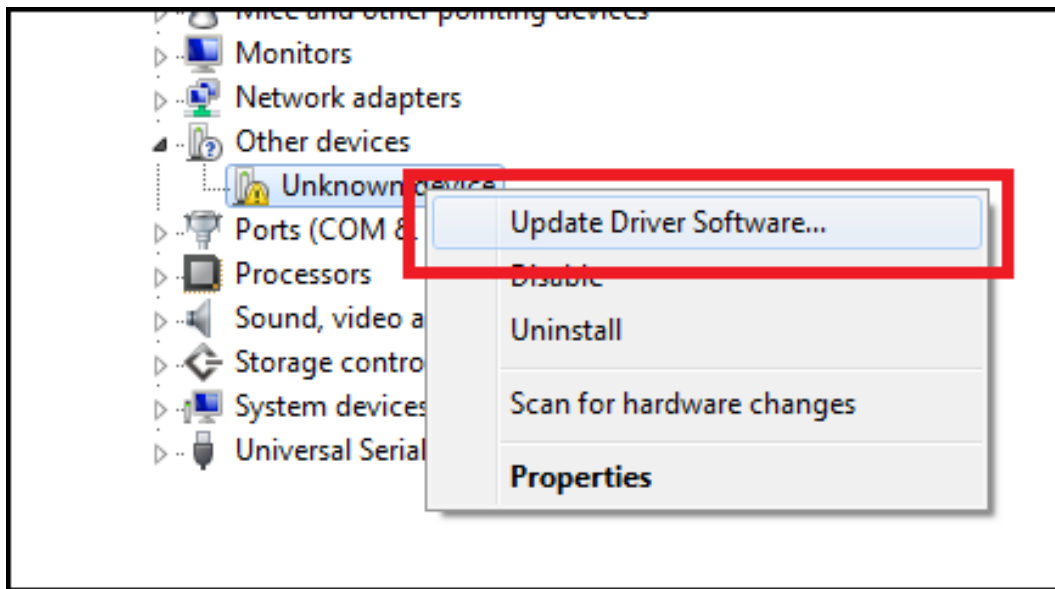


Select device manager from left side.

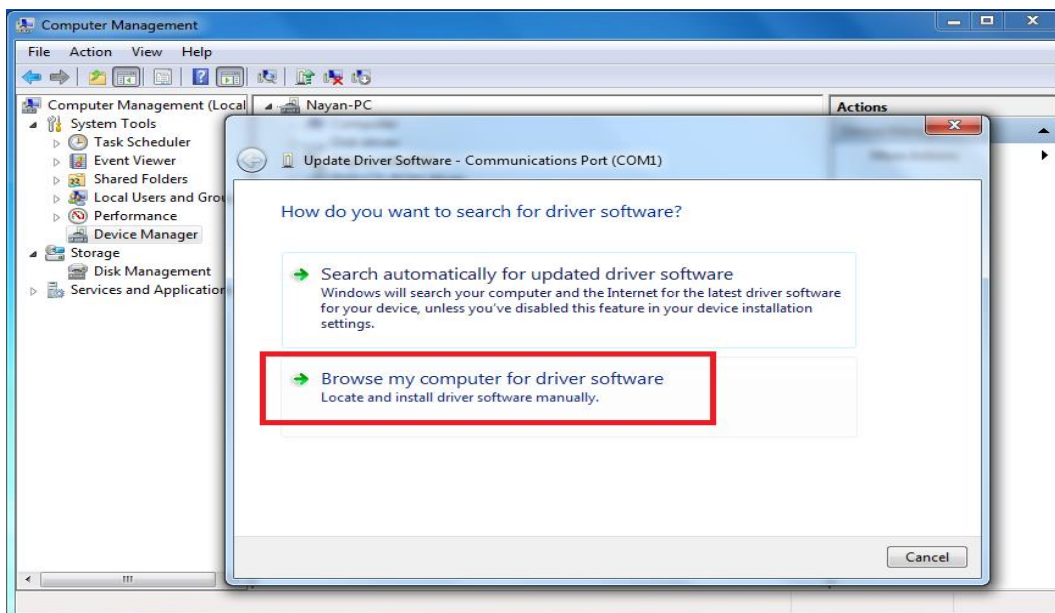
At right side an unknown device can be seen as shown below;



Right click on the **Unknown device** and select '**Update Driver Software**'.



Select '**Browse my computer for driver software**'.



Click on '**Browse**' and select the path for FTDI Drivers according to the Operating system discussed earlier. The path may be anywhere the FTDI drivers has been saved.

Here, the operating system is windows 7 (32-bit), so the path is,

D:\FTDI Drivers\Windows-7 (32- bit)

Select the path and click on **NEXT**.



After completion, you can find a specific COM port in Device manager under 'Ports (COM & LPT)', here, it is **COM-13**.



Now, the FOL-CWDM device is ready to be used and operated using PC.

IV. CHECK LIST FOR INSTALLED ITEMS:

Check whether all the files are properly installed or not.

Considering default path as;

C:\Program Files\RFL-RFGD, the item list is given below,



If these files are missing, re install the software.

Again go to,

C:\Program Files\National Instruments\shared

Check for the following folders are present or not,

- Lab VIEW Run Time
- NI-VISA

If there folder are absent, re install the software.

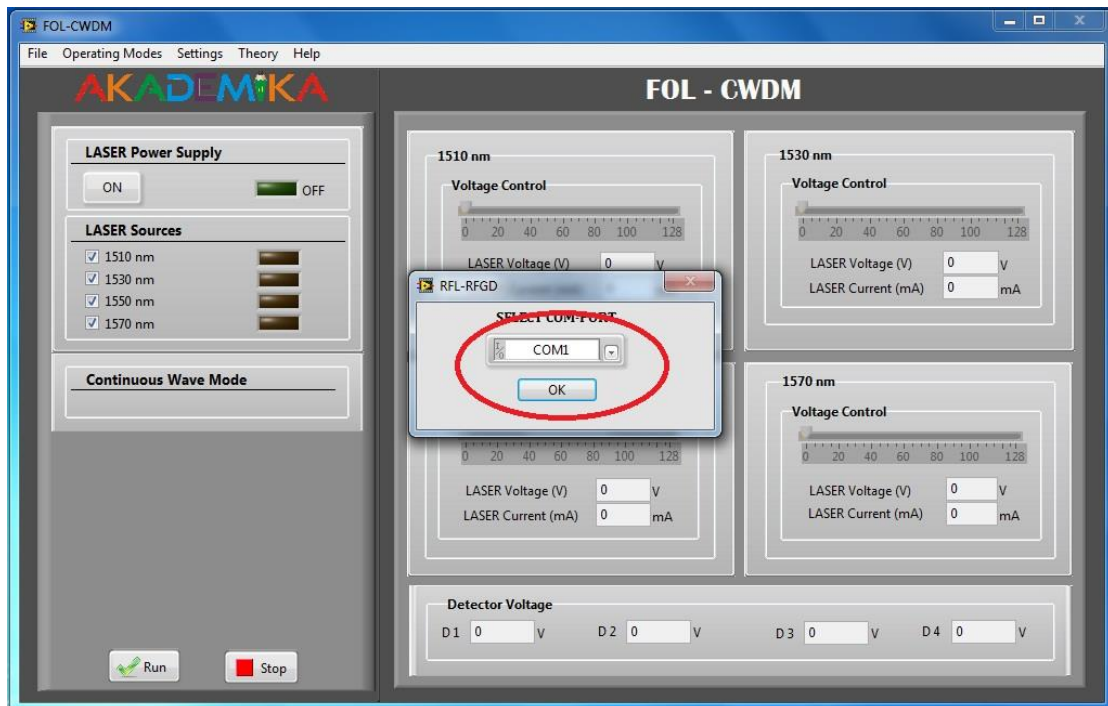
TURNING ON THE KIT & SOFTWARE OPERATION

TURNING ON THE KIT:

- 1 Refer to the back panel diagram from Back Panel Details Section.
- 2 Connect the Power Chord to the Power Connector.
- 3 Refer to the below figure to make connections from the source to the detector using SC/PC – SC/PC single mode fiber optic patch chords provided along with the system.



- Switch ON the system from the back panel.
- Connect Shielded USB A-B cable from system to the PC.
- Check if the USB connection is detected at PC end.
- Run the FOL CWDM.exe file from installed directory.
(Default directory **C:\Program Files\CWDM\FOL CWDM.exe**)
- When asked for the COM-Port, select the appropriate COM-Port assigned by the driver and click 'OK'.



- At this point system is in Continuous mode by default. This allows individual LASER to be turned ON, set its forward current and view the Laser forward current, forward voltage as well as detector output (in voltage) in display window.
- To turn ON the system switch ON the laser power supply by clicking 'ON' button from Laser Power Supply section. Green light indicates supply is turned ON.



- To turn ON individual Laser diodes select the appropriate checkbox corresponding to laser wavelength and click the 'RUN' button.

- By default voltage control sliders from individual LASER SECTIONS are at minimum position. These sliders allow setting the control voltage to adjust the forward current from individual laser diodes. Sliders are accessible only after RUN button is clicked.
- Adjust the sliders to set the forward current of the laser diode by observing current value in 'Current (mA)' field.
- As the current varies the forward voltage across the laser diode is displayed in 'Voltage (V)' field.
- Forward current in the laser diode sets the output power which is measured by the photo-detector. The photo-detector output in voltage is displayed in 'Detector Voltage' section.

POWER MEASUREMENT USING OPTICAL POWER METER

- Instead of connecting source output to the detector use Fiber Optic Power Meter at the other end of the patch cord.
- Select the wavelength 1550nm on Power Meter and Power scale as Linear (Watt).
- Using voltage control slider vary the forward current and measure the output optical power for individual Laser sources.



OPERATION MODES

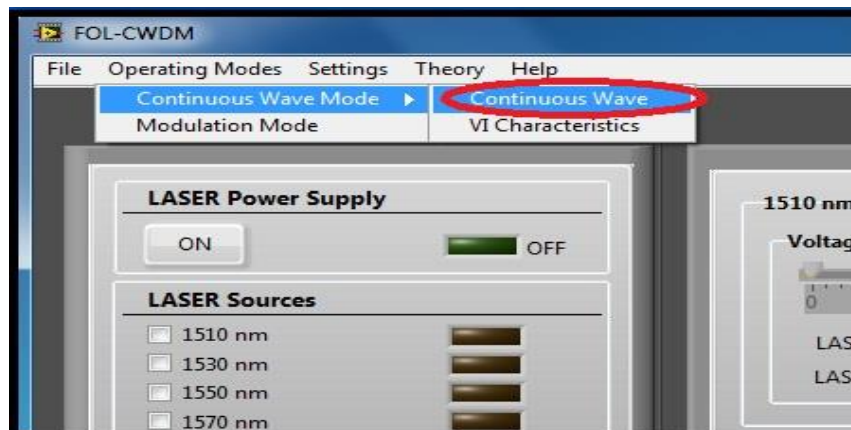
1. Continuous Wave Mode :

The Continuous wave mode is further divided into two categories; Continuous wave (CW) and VI Characteristics.

i. Continuous Wave (CW):

This is the default operation mode. To navigate to the continuous mode, go to;

Menu bar >> Operation Modes >> Continuous wave mode >> Continuous wave



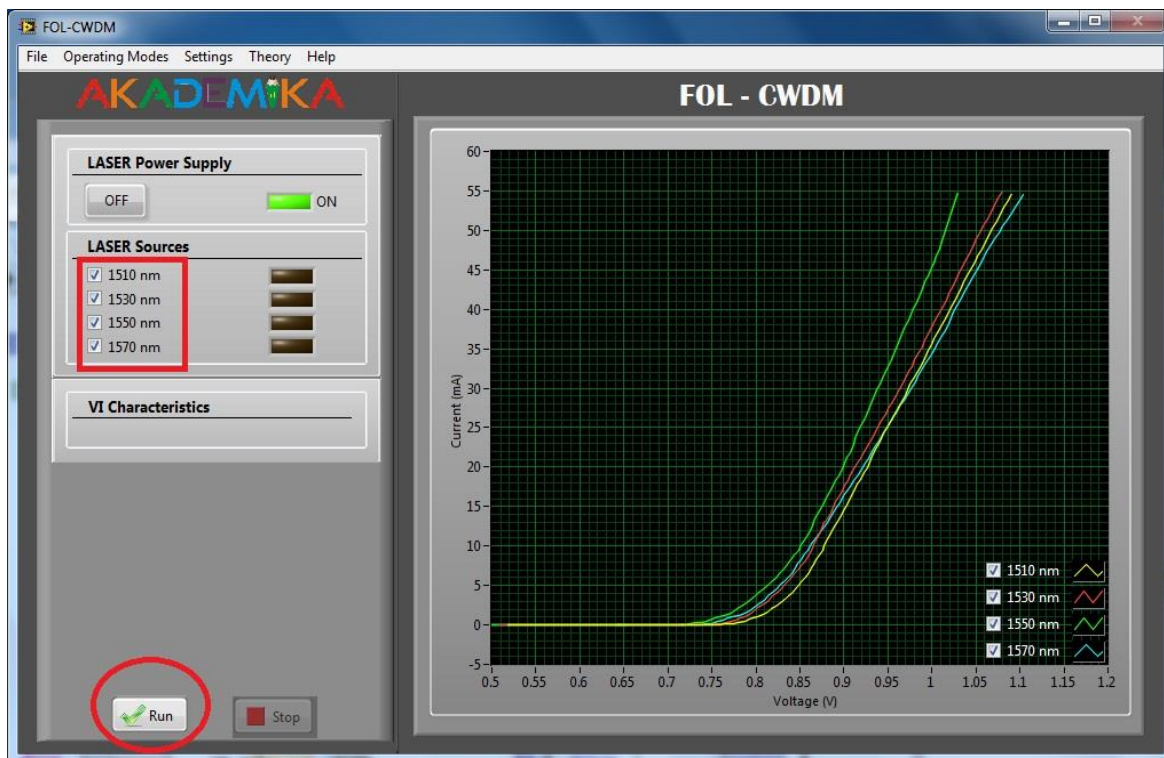
The CW mode operation is same as explained in 'Turning ON the Kit' section. Here the laser diodes are not modulated but generate constant power.

ii. VI characteristics :

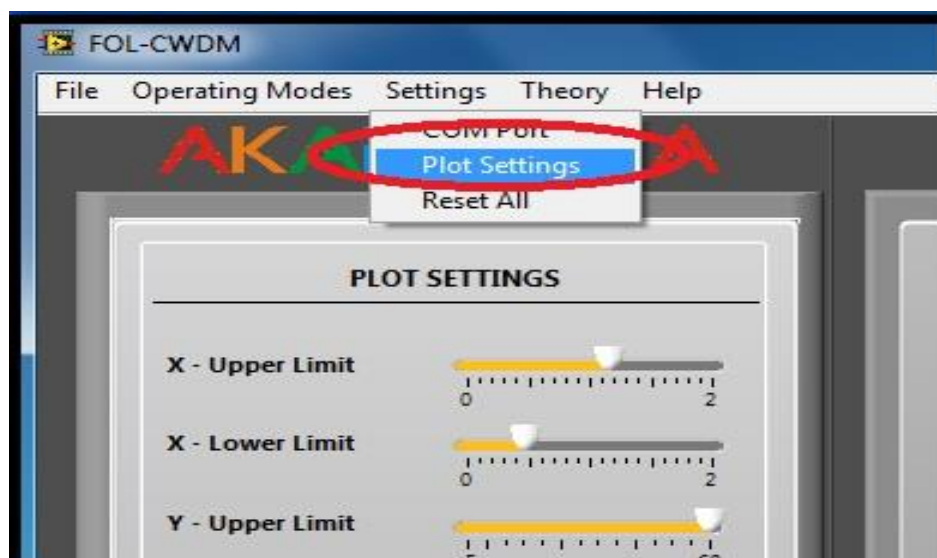
VI characteristics mode is invoked by selecting the options in following sequence.

Menu bar >> Operation Modes >> Continuous wave mode >>VI Characteristics

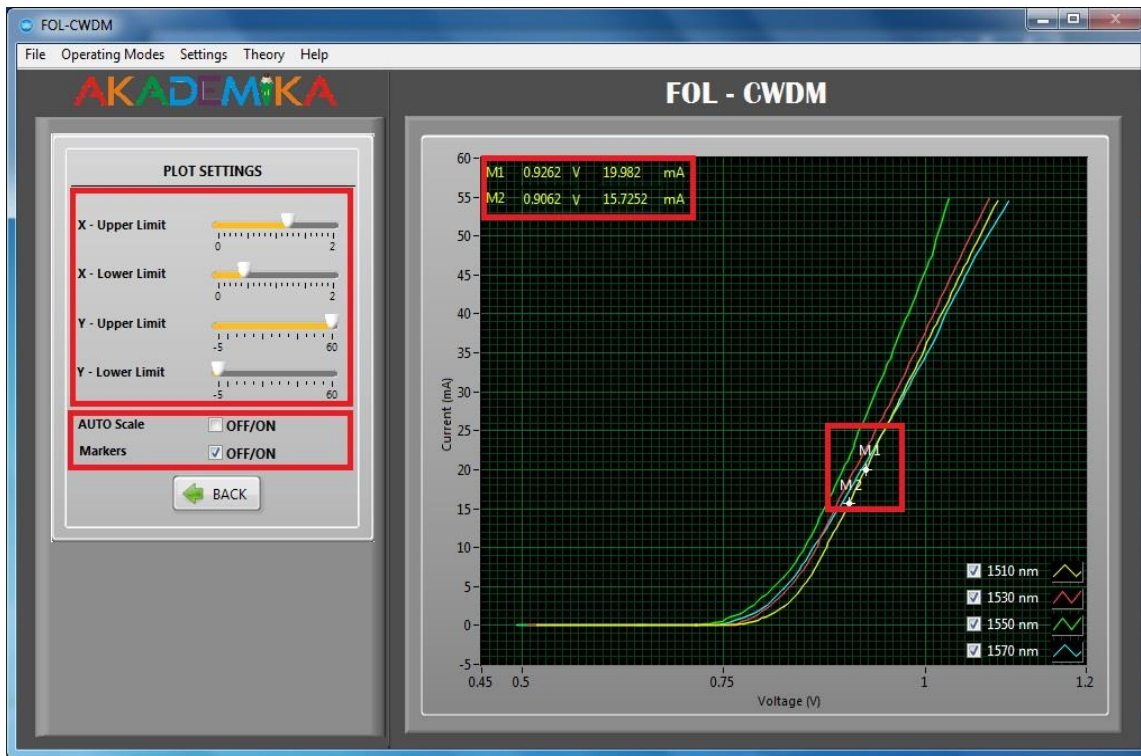
To plot the vi characteristics select the laser sources and click the RUN button.
The VI characteristics can be seen on the graph as shown below.



To set the graph scales, go to;
Menu bar >> Settings >> Plot Settings

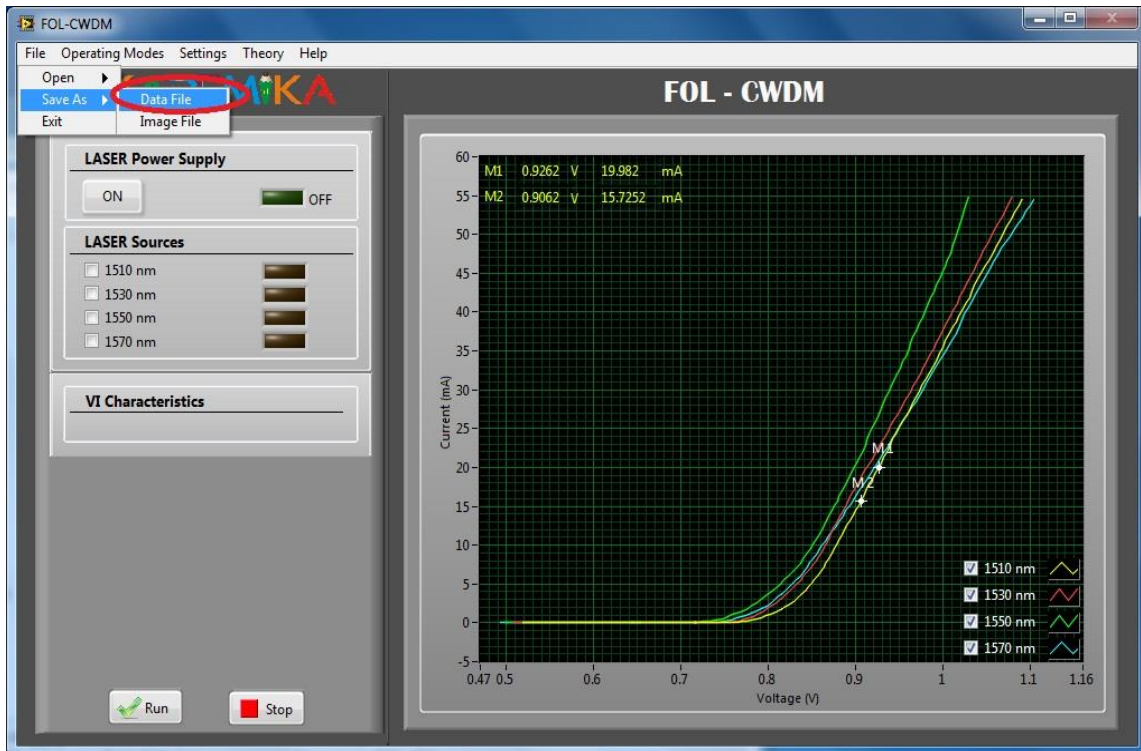


Adjust the X-Y upper and lower limits, turn markers ON/OFF etc. as shown in the fig.

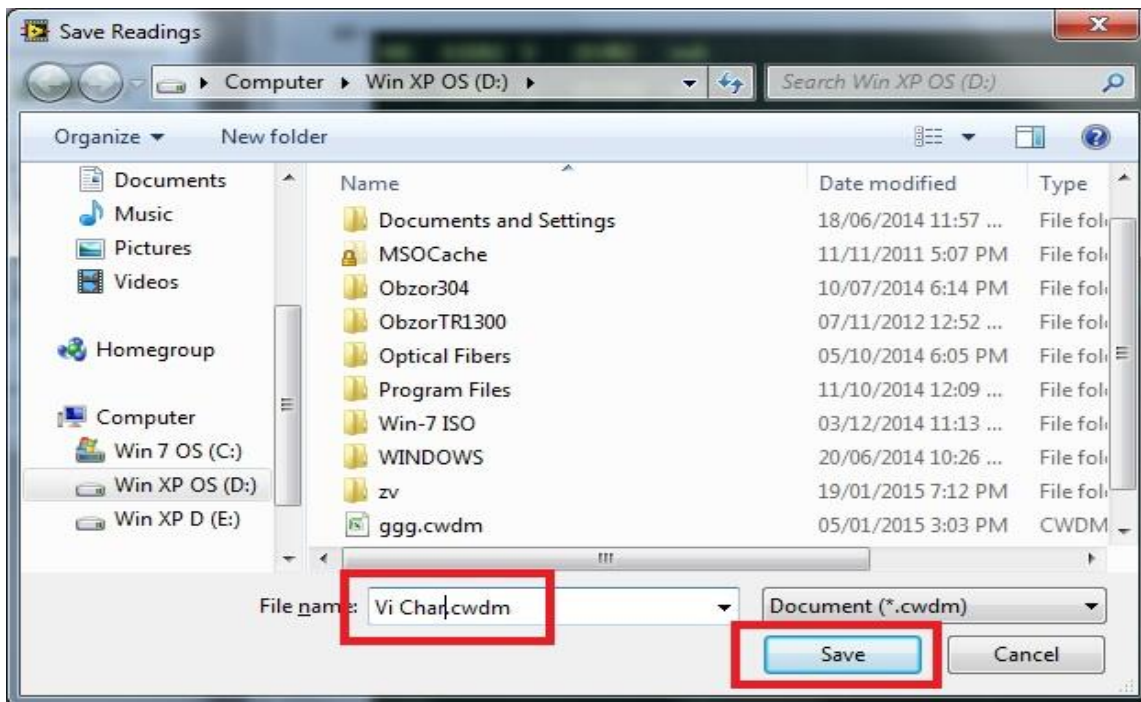


To save the readings, Go to;

Menu bar >> File >> Save As >> Data File

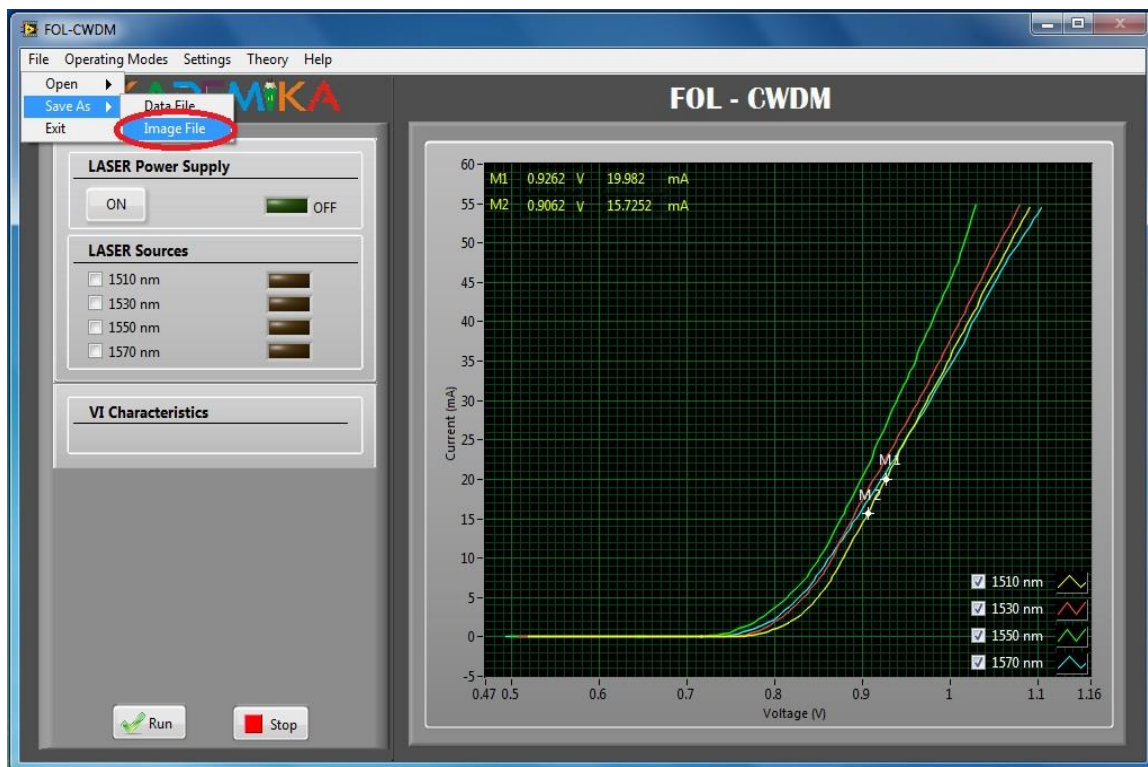


The extension of the file would be (*.cwm).

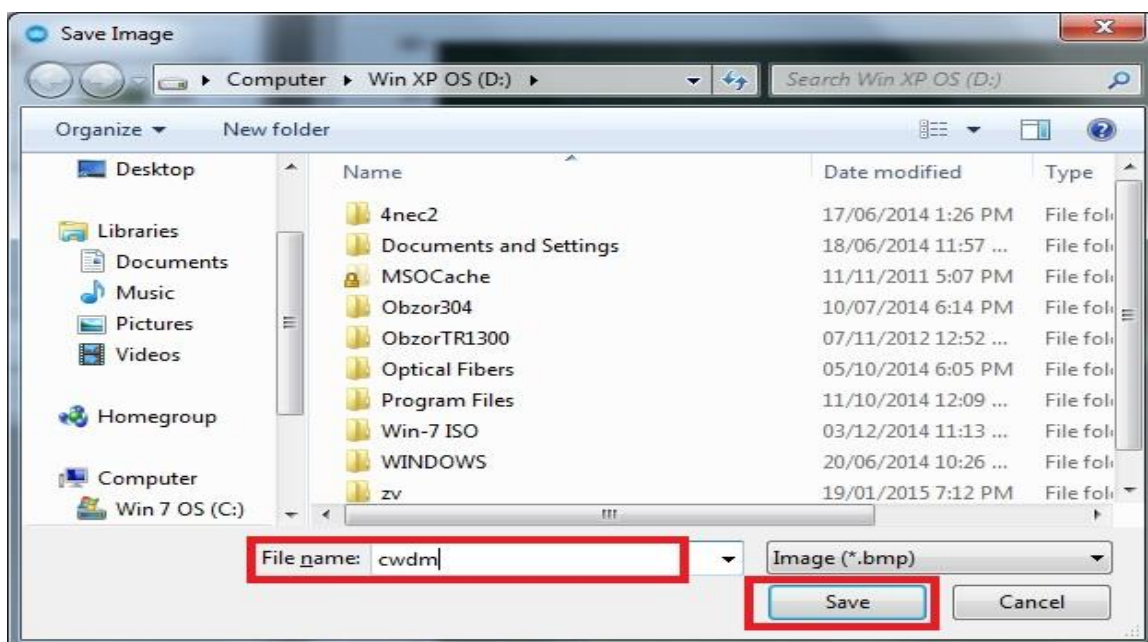


To save the graph image, go to;

Menu bar>> File >> Save As >>Image File



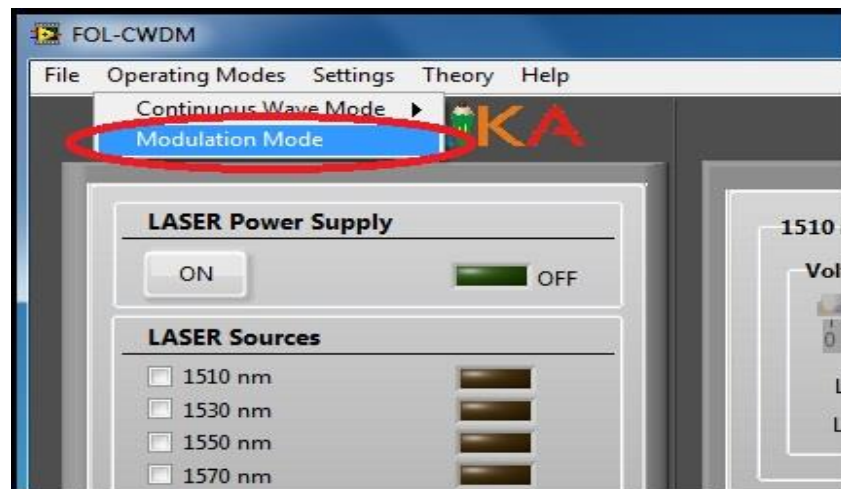
The extension of the image file would be (*.bmp)



2. Modulation Mode:

To navigate to the modulation mode, go to;

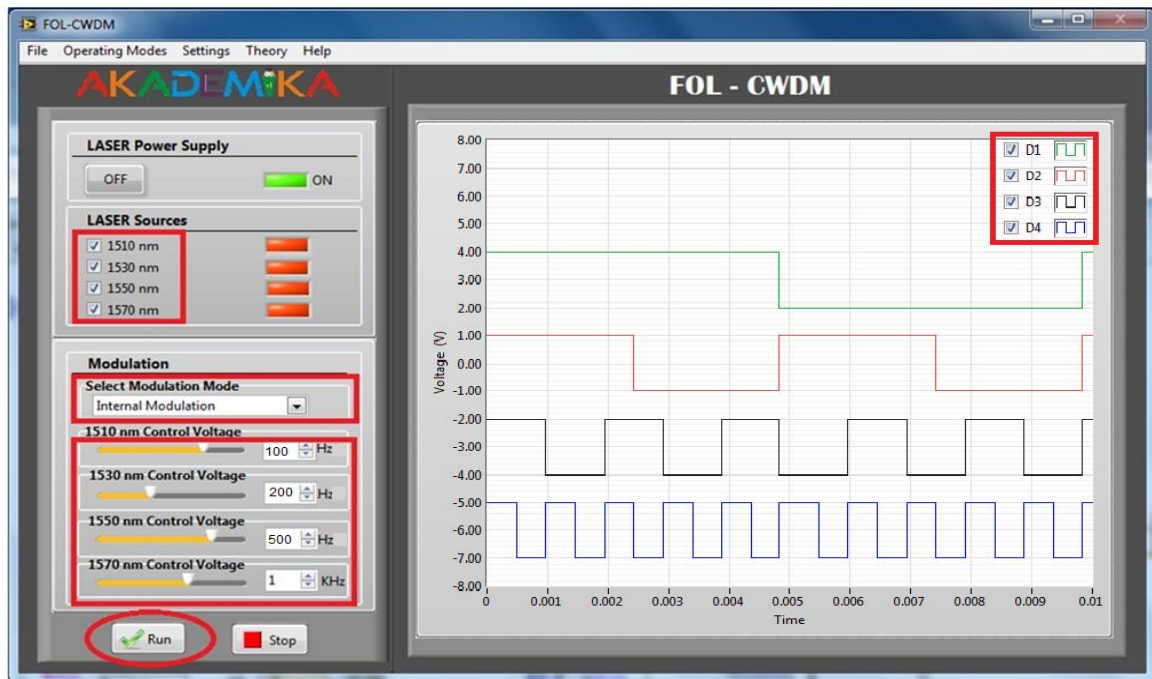
Menu bar >> Operation modes >> Modulation



At the device end, connect the Detector O/P (electrical) externally to oscilloscope.

i. Internal Modulation :

- Select 'Internal Modulation' in Modulation mode selector dropdown.
- To turn ON individual Laser diodes select the appropriate checkbox corresponding to laser wavelength.
- Set the control voltage sliders and modulation frequency among 100Hz, 200Hz, 500Hz and 1 KHz for each of the sources and click the 'RUN' Button.
- Observe the modulated signals on the graph. These modulated signals can also be seen on the oscilloscope externally by connecting the electrical detector output to oscilloscope by using BNC to BNC cable.
- Click 'STOP' button to stop the experiment.

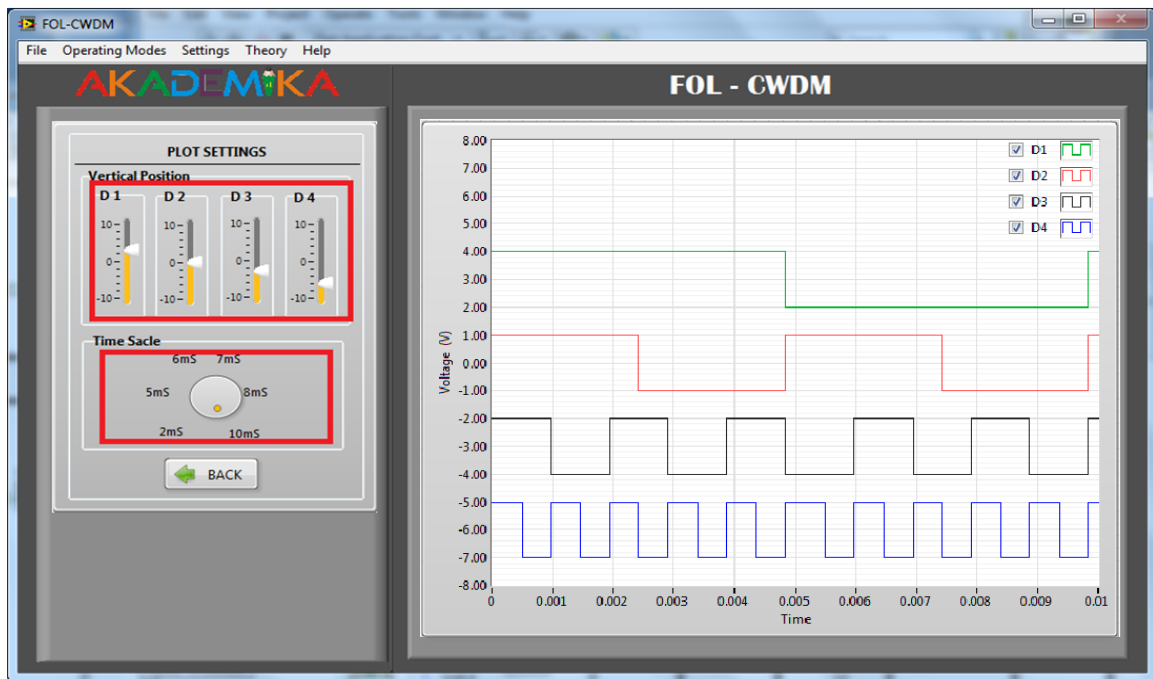


To adjust the Time Scale and compare all of the four plots, go to;

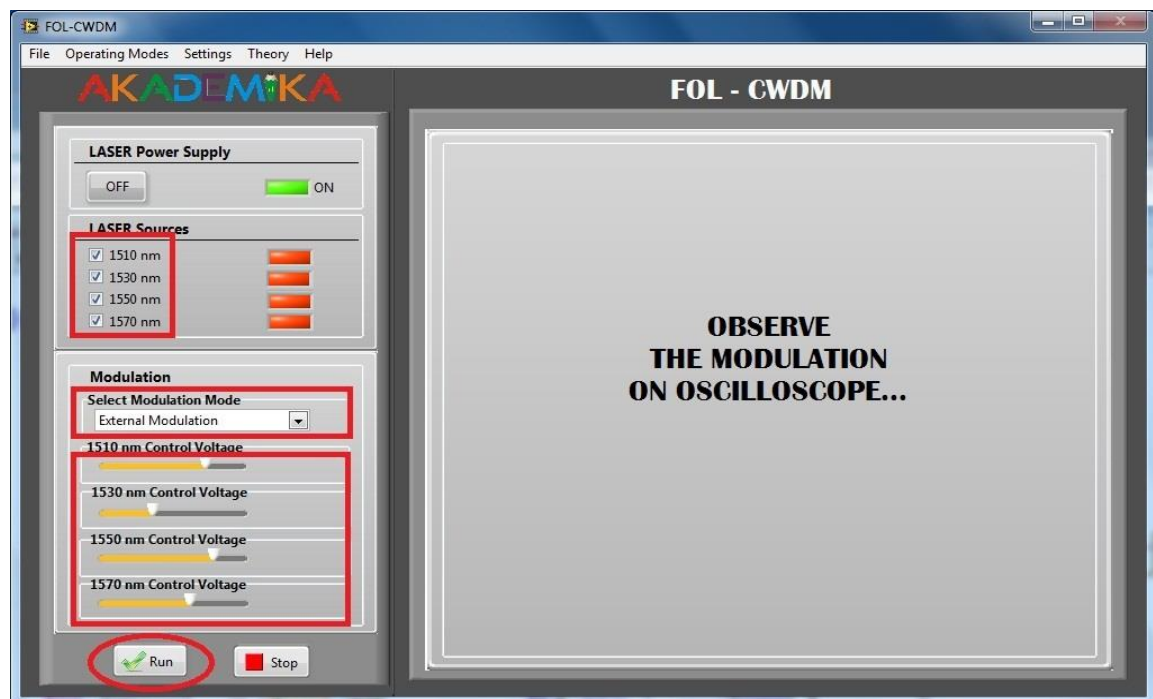
Menu bar >> Settings >> Plot settings

Adjust Vertical position of the plots D1, D2, D3 D4 by using vertical sliders D1, D2, D3 and D4 respectively as shown in the figure.

Adjust the time base by using the Time scale knob.



ii. External Modulation :



- Select 'External Modulation' in Modulation mode selector dropdown.

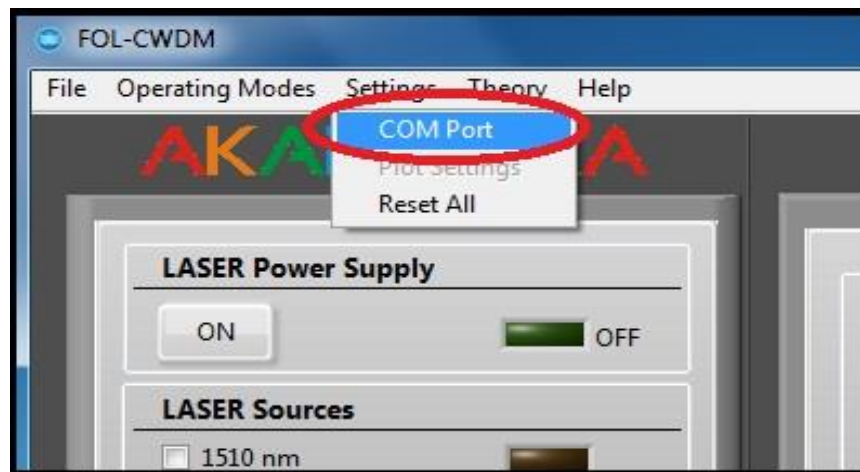
- To turn ON individual Laser diodes select the appropriate checkbox corresponding to laser wavelength, adjust the control voltage sliders for each source and click the 'RUN' button.
- Connect the BNC cables coming from the external Function Generator to the External I/Os on the back panel of the FOL-CWDM kit.
- Connect BNC to BNC cable from electrical detectors D1, D2, D3 and D4 to the Oscilloscope.
- The output signals will be seen on the Oscilloscope as shown below.
- Click 'STOP' button to stop the modulation.

SETTINGS:

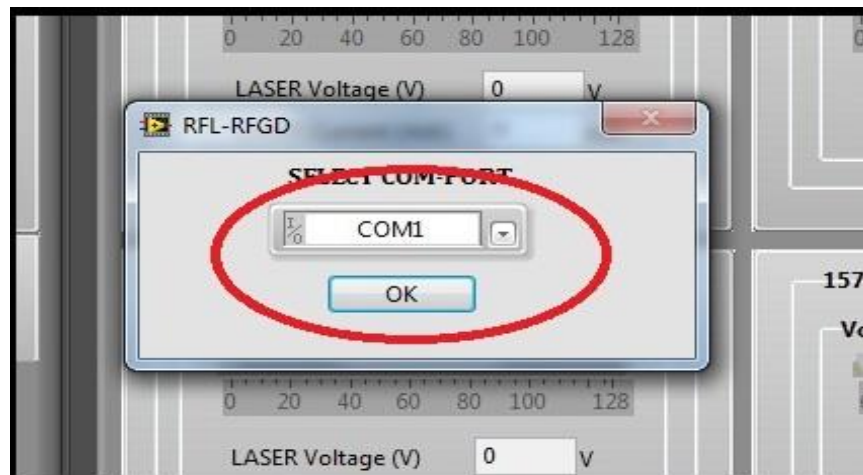
1. COM-Port Settings

To set the COM-Port, navigate to;

Menu bar >> Settings >> COM Port



Select the **COM-Port** and click on **OK**.

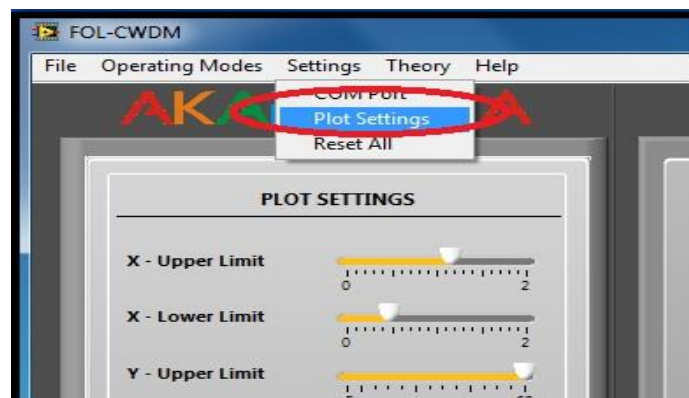


2. Plot Settings

To change the plot settings, go to;

Menu bar << Settings << Plot Settings

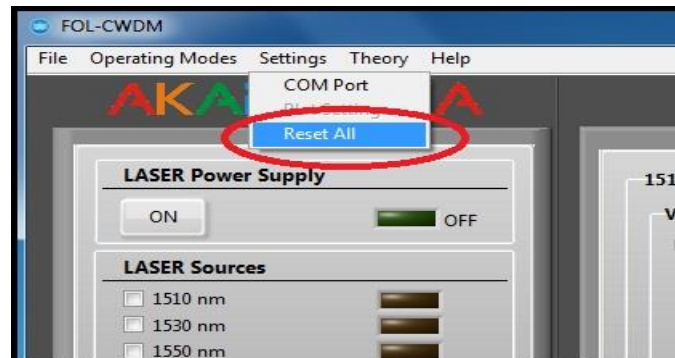
The respective plot parameters can be changed here.



3. Reset All:

To reset the system, navigate to;

Menu bar >> Settings >> Reset All



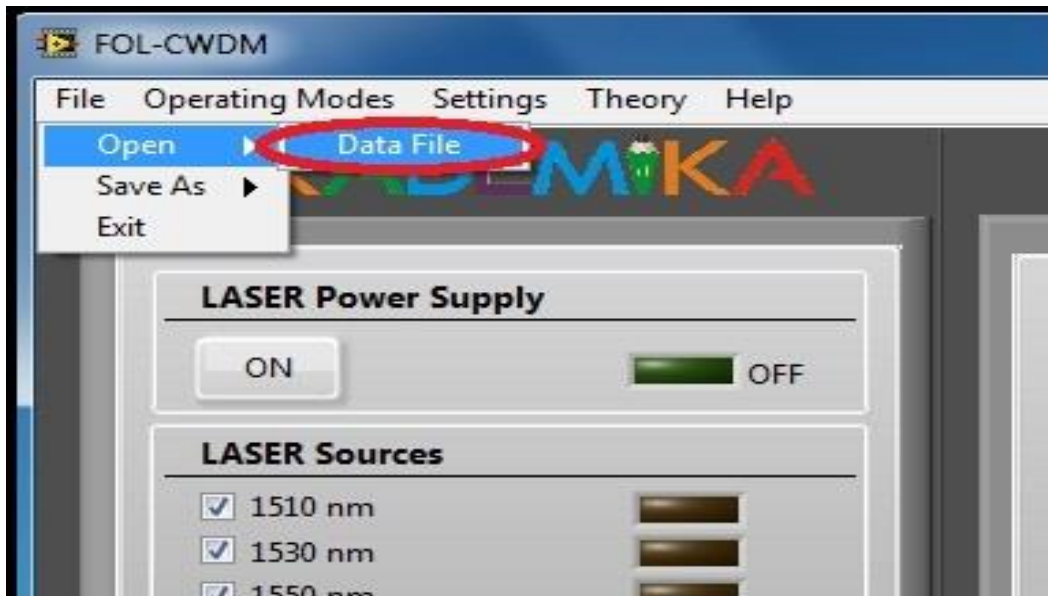
A popup will appear showing that the device is reset.



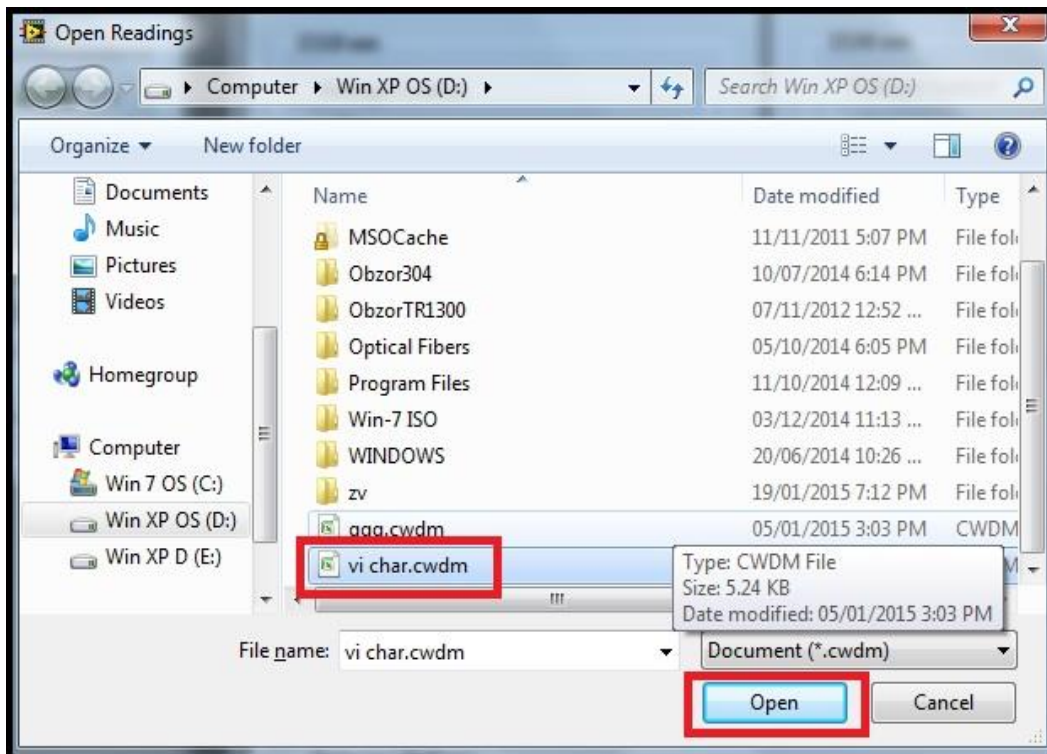
V. OPEN DATA FILE :

A previously saved graph can also be opened. Go to,

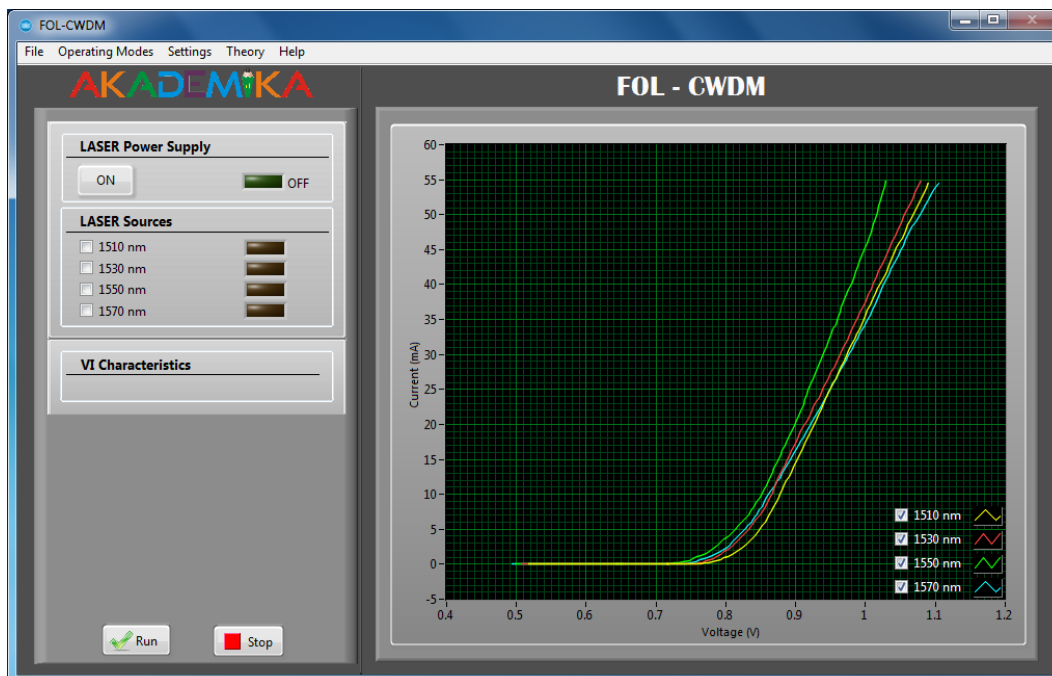
Menu Bar << File << Open << Data file



Choose the data file to open.



The graph will be plotted.



VI. Theory :

Go to;

Menu bar >>Theory

Select any of the topics available to open *.pdf document.

VII. Help :

Go to

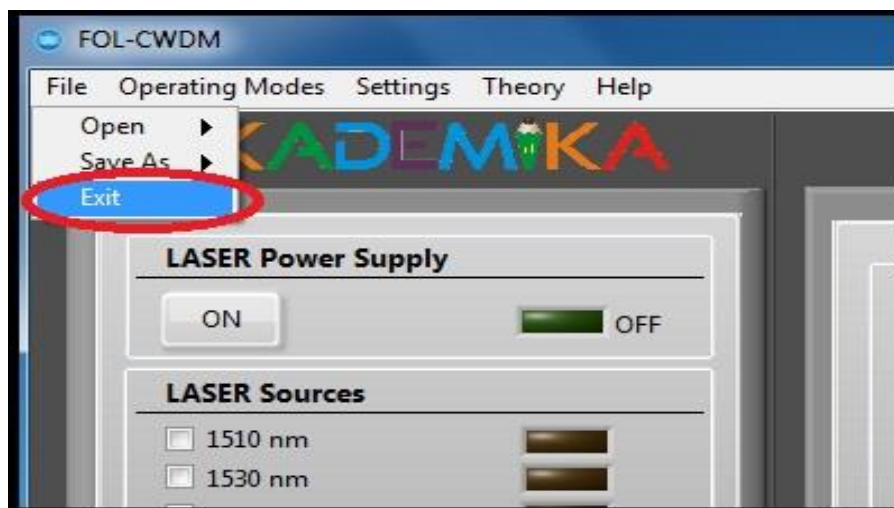
Menu bar >> Help

Select any of the topics available to open *.pdf document.

VIII. EXIT :

Exit from the software, go to;

Menu bar << File << Exit



The software will be closed.

EXPERIMENT NO.1

EXPERIMENT NO : 1

NAME : -

Continuous Wave Operation and IP characteristics.

OBJECTIVE : -

To understand CW operation of the laser diodes and plot IP characteristics.

APPARATUS : -

FOL – CWDM Trainer Kit.

Power cord.

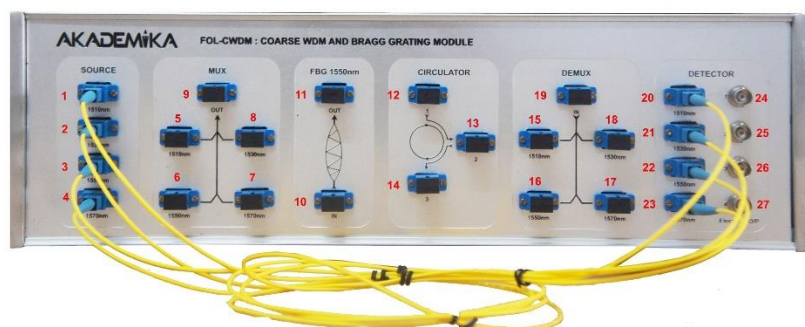
USB A to B cable.

SC/PC – SC/PC fiber patch cords.

Optical Power Meter.

PROCEDURE:

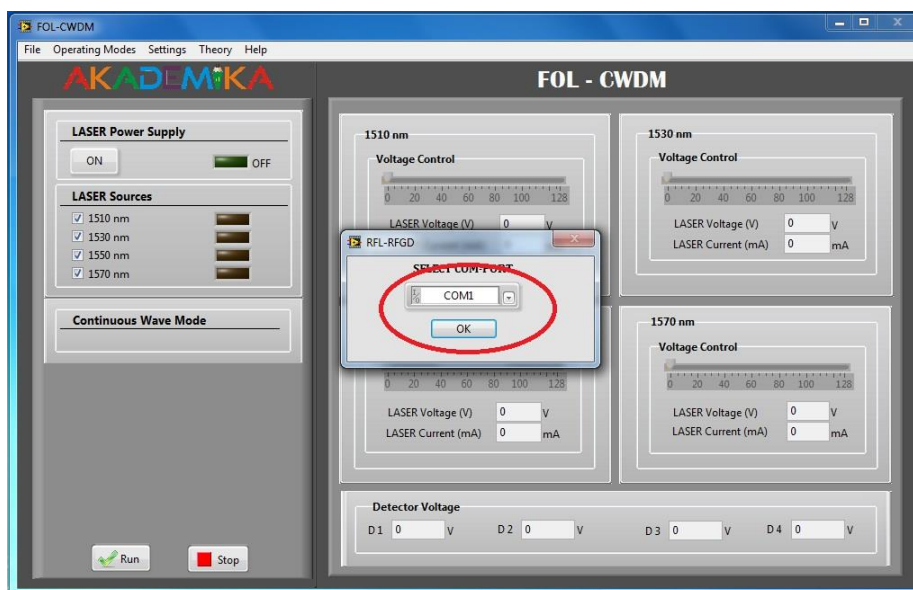
1. Refer to the back panel diagram from Back Panel Details Section.
2. Connect the Power Chord to the Power Connector.
3. Refer to the below figure to make connections from the source to the detector using SC/PC – SC/PC single mode fiber optic patch chords provided along with the system.



4. Switch ON the system from the back panel.
5. Connect Shielded USB A-B cable from system to the PC.
6. Check if the USB connection is detected at PC end.
7. Run the FOL CWDM.exe file from installed directory.

(Default directory **C:\Program Files\CWDM\FOL CWDM.exe**)

8. When asked for the COM-Port, select the appropriate COM-Port assigned by the driver and click 'OK'.



9. At this point system is in Continuous mode by default. This allows individual LASER to be turned ON, set its forward current and view the Laser forward current, forward voltage as well as detector output (in voltage) in display window.
10. To turn ON the system switch ON the laser power supply by clicking 'ON' button from Laser Power Supply section. Green light indicates supply is turned ON.



11. To turn ON individual Laser diodes select the appropriate checkbox corresponding to laser wavelength and click the 'RUN' button.

12. By default voltage control sliders from individual LASER SECTIONS are at minimum position. These sliders allow setting the control voltage to adjust the forward current from individual laser diodes. Sliders are accessible only after RUN button is clicked.
13. Adjust the sliders to set the forward current of the laser diode by observing current value in 'Current (mA)' field.
14. As the current varies the forward voltage across the laser diode is displayed in 'Voltage (V)' field.
15. Forward current in the laser diode sets the output power which is measured by the photo-detector. The photo-detector output in voltage is displayed in 'Detector Voltage' section.

POWER MEASUREMENT USING OPTICAL POWER METER

- 1 Instead of connecting source output to the detector use Fiber Optic Power Meter at the other end of the patch chord.
- 2 Select the wavelength 1550nm on Power Meter and Power scale as Linear (Watt).
- 3 Using voltage control slider vary the forward current and measure the output optical power for individual Laser sources.
- 4 Plot the graph of Forward Laser Current verses Optical Output Power.



OBSERVATIONS:

Thus, we studied CW operation and IP characteristics of the laser diodes.

EXPERIMENT NO.2

EXPERIMENT NO : 2

NAME:

To determine insertion loss of each channel of WDM Mux

To determine the optical cross talk in adjacent channels of WDM Demux for various wavelengths

EQUIPMENTS :

1. C-Band Lasers - 4
2. 4 channel WDM Mux
3. 4 channel WDM Demux
4. Photodetectors - 4

THEORY:

Insertion Loss = $-10 \log (P_{out}/P_{in})$

P_{in} is the transmitted optical power without Mux and P_{out} is the transmitted power with Mux.

Cross talk = $10 \log (P_j/P_i)$

P_i is the optical output power at the specific wavelength designated for the port i , and P_j is the output power at the same port (i) for adjacent/other channel wavelengths.

WDM Mux and Demux

A WDM mux is a device that combines a set of different wavelengths propagating in different fibers connected at the input ports of the device into one output fiber. Similarly a WDM Demux is a device that separates different wavelengths propagating through a single fiber into separate fibers. These devices are very important components in a wavelength division multiplexed fiber optic communication system. In such a system, different wavelengths carry independent information and all the wavelengths use the same fiber to propagate from one terminal to the other. Thus they all use the same transmission fiber which results in an increased capacity of the link. Thus if there are 4 wavelength channels each carrying bit rates of 2.5 Gb/s, then when multiplexed into the fiber link the fiber link would be carrying a total

capacity of $4 \times 2.5 = 10$ Gb/s. Larger the number of multiplexed channels larger would be the total capacity of the communication link.

WDM began in the late 1980s using the two widely spaced wavelengths in the 1310 nm and 1550 nm (or 850 nm and 1310 nm) regions, sometimes called wideband WDM. The early 1990s saw a second generation of WDM, sometimes called narrowband WDM, in which two to eight channels were used. These channels were now spaced at an interval of about 400 GHz in the 1550-nm window. By the mid-1990s, dense WDM (CWDM) systems were emerging with 16 to 40 channels and spacing from 100 to 200 GHz. By the late 1990s CWDM systems had evolved to the point where they were capable of 64 to 160 parallel channels, densely packed at 50 or even 25 GHz intervals.

A WDM mux or demux is characterized by some important operating characteristics. These include:

a. Insertion loss: If P_1 is the power entering the mux at a specific wavelength and P_2 is the power exiting the mux then the insertion loss of the mux at this wavelength is defined by

$$IL(dB) = -10 \log \left(\frac{P_2}{P_1} \right)$$

Thus if for an input power of 1 mW the output power is 0.8 mW then the insertion loss would be 0.97 dB.

Insertion loss is a very important characteristic of a multiplexer or a demultiplexer and specifies the loss encountered by signals passing through the multiplexer or the demultiplexer. The lower the insertion loss, the better is the device. The loss encountered by the signals at the multiplexer and demultiplexer should be factored in the design of the fiber optic link.

b. Cross talk: It refers to how well two different wavelength channels are isolated in a given output. Thus if the output port 1 is designated for the wavelength λ_1 is to exit, in principle there should be no light at any other wavelength exiting from port 1. Due to the design or fabrication errors, a small amount of power at other wavelengths (usually corresponding to adjacent channels) may also be exiting from the same output port. This is referred to as cross talk.

Let λ_i be the wavelength which is supposed to exit from the i th port of a demux. If P_i is the power exiting at P_i from the i th port of the demux and if P_j is the power exiting the i th port at any other wavelength P_j then we define the cross talk of the i th output as

$$CT(dB) = 10 \log \left(\frac{P_j}{P_i} \right)$$

If the j th channel is an adjacent channel to the i th channel then the cross talk is referred to as adjacent channel cross talk; otherwise it is referred to as non-adjacent channel cross talk.

For example, let us consider a four wavelength demultiplexer. Assume that at the input of the demux equal powers of 1.2 mW at each wavelength λ_1 to λ_4 are incident. If the output from Port # 1 (the intended port for wavelength λ_1) at a wavelength λ_1 is 1 mW and if at the same output port the powers at λ_2 , λ_3 , and λ_4 are 0.02 mW, 0.01mW and 0.005 mW, respectively, then the cross talk at the Port 1 comes out to be -17 dB adjacent channel cross talk, and -20 dB and -23 dB non adjacent channel cross talks respectively.

It may be mentioned that, apart from errors in the design or fabrication of the device, any deviation of the wavelength of the laser source can also result in deterioration of the performance of the multiplexer or the demultiplexer in terms of insertion loss and cross talk.

PROCEDURE :

For Insertion Loss

1. Take out the optical power from one of the C-Band Lasers, say 1510nm laser, using a patch cord and measure using one of the InGaAs photodetectors, say PD1.
2. Adjust The laser power for operation below detector saturation (4V) and is measure as P1.
3. Remove the patch cord end connected to the detector and connect to the 1510nm channel of Mux.
4. Feed the optical power from the OUTPUT port of Mux to the same photodetector, PD1 using another patch cord and measure the optical power as P2.
5. The above given formula is used to find out the insertion loss for that channel. The insertion loss should be corrected for 0.5 dB extra connector loss due to the additional patch cord used to include the Mux in the light path.
6. Repeat the measurement for other channels using other C-Band Lasers (one at a time) and any of the InGaAs photodetectors.

For CrossTalk

1. Connect one of the C-Band Lasers, say 1550nm laser, to the COM port of Demux using a patch cord and connect its 1550nm output channel (port i) to the optical power meter using another patch cord.
2. Measure the optical power as P_i .
3. Similarly measure the optical power, P_j coming at other ports (1510, 1530 and 1570) also .
4. Use the formula to find out the optical cross talk.
5. Repeat the experiment using other C-Band Lasers.

OBSERVATION:

Insertion Loss of Mux Channels

Channel Wavelength (nm)	Optical power without mux, P_{in} (mW)	Optical power with mux, P_{out} (mW)	Insertion Loss in dB= - $10 \log (P_{out}/P_{in})$
1510			
1530			
1550			
1570			

Measurements for Optical Cross Talk

Laser Wavelength (nm)	1510nm channel, P_1 (mW)	1530nm channel, P_2 (mW)	1550nm channel, P_3 (mW)	1570nm channel, P_4 (mW)
1510				
1530				
1550				
1570				

Cross talk for 1510nm:

Cross talk at 1530 channel = $10 \log (P_2/P_1) =$

Cross talk at 1550 channel = $10 \log (P_3/P_1) =$

Cross talk at 1570 channel = $10 \log (P_4/P_1) =$

Cross talk for 1530nm:

Cross talk at 1510 channel = $10 \log (P_1/P_2) =$

Cross talk at 1550 channel = $10 \log (P_3/P_2) =$

Cross talk at 1570 channel = $10 \log (P_4/P_2) =$

Cross talk for 1550nm:

Cross talk at 1510 channel = $10 \log (P_1/P_3) =$

Cross talk at 1530 channel = $10 \log (P_2/P_3) =$

Cross talk at 1570 channel = $10 \log (P_4/P_3) =$

Cross talk for 1570nm:

Cross talk at 1510 channel = $10 \log (P_1/P_4) =$

Cross talk at 1530 channel = $10 \log (P_2/P_4) =$

Cross talk at 1550 channel = $10 \log (P_3/P_4) =$

RESULT :

The insertion loss of each WDM Mux channel and the loss uniformity is determined.

The optical cross talk involved in WDM Demux is measured.

EXPERIMENT NO.3

EXPERIMENT NO : 3

NAME :

IV characteristics of LASER Diode.

OBJECTIVE :

To plot IV characteristics of LASER Diode.

APPARATUS :

FOL – CWDM Trainer Kit.

Power cord.

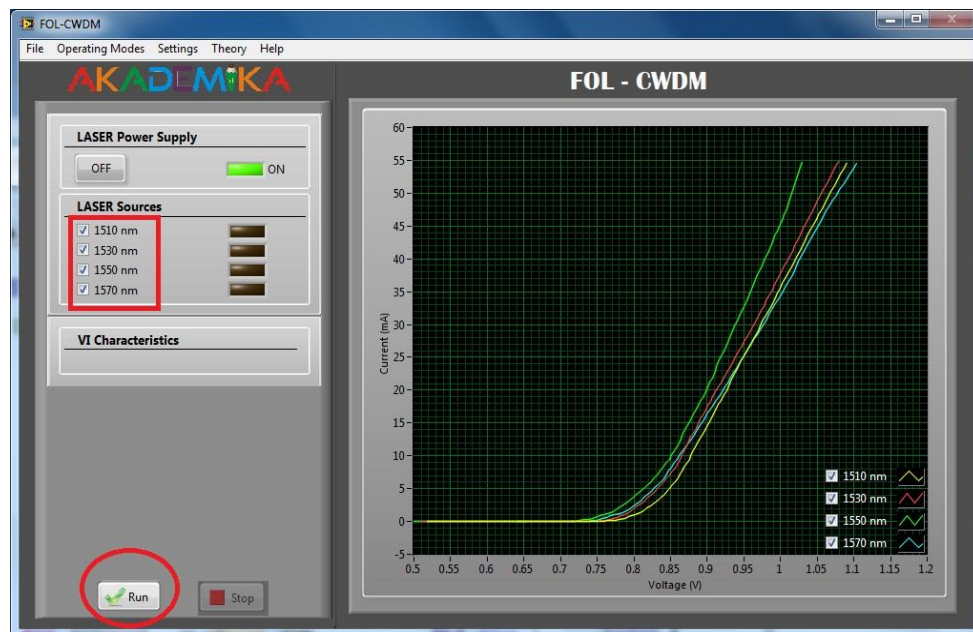
USB A to B cable.

PROCEDURE :

1. Setup the FOL-CWDM kit as explained in experiment 1 except the connectivity of the source to the detector by fiber optic patch cords.
2. VI characteristics mode is invoked by selecting the options in following sequence.

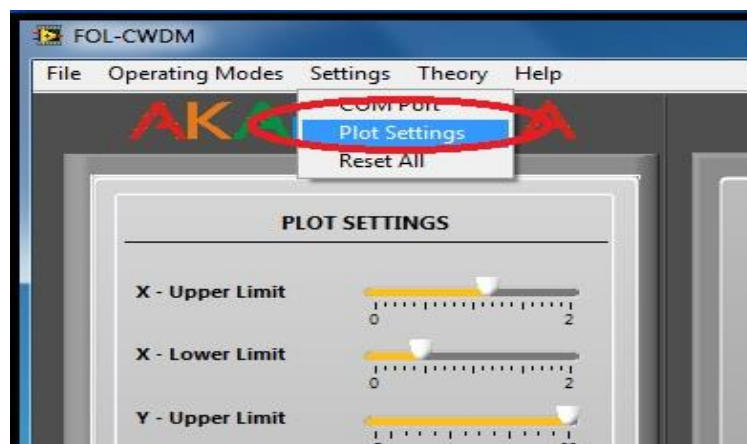
Menu bar >> Operation Modes >> Continuous wave mode >>VI Characteristics.

3. To plot the vi characteristics select the laser sources and click the RUN button.
4. The VI characteristics can be seen on the graph as shown below.

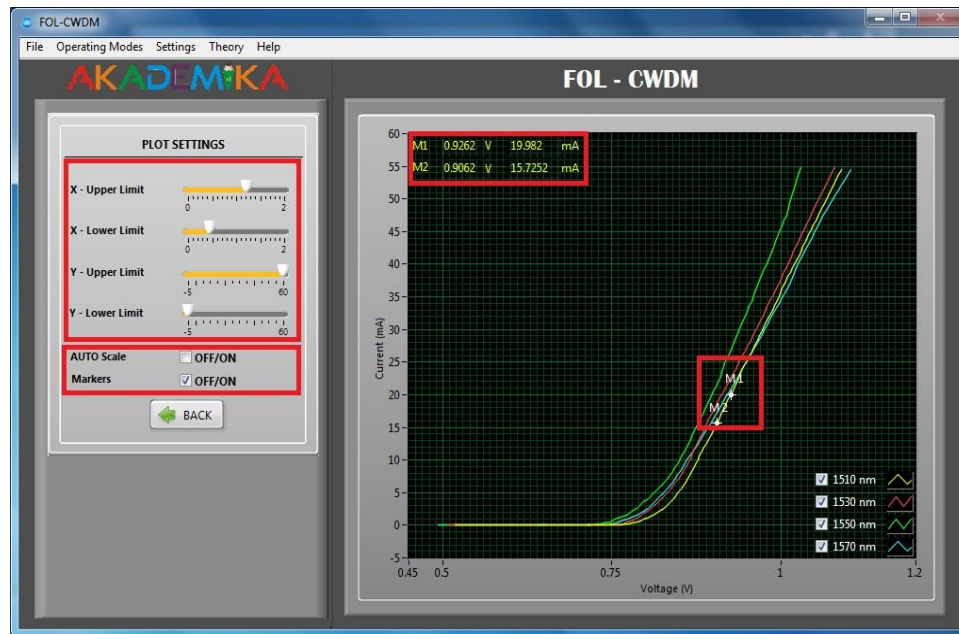


5. To set the graph scales, go to;

Menu bar >> Settings >> Plot Settings



6. Adjust the X-Y upper and lower limits, turn markers ON/OFF etc. as shown in the fig.



OBSERVATIONS :

In this experiment we studied IV characteristics of LASER Diode and plotted Graph for the same.

EXPERIMENT NO.4

EXPERIMENT NO : 4

NAME : -

Forming communication link using digital modulation.

OBJECTIVE : -

To develop communication link using Internal and External modulation.

APPARATUS : -

FOL – CWDM Trainer Kit.

Power chord.

USB A to B cable.

SC/PC – SC/PC fiber patch cords.

Fiber Optic Power Meter.

DSO.

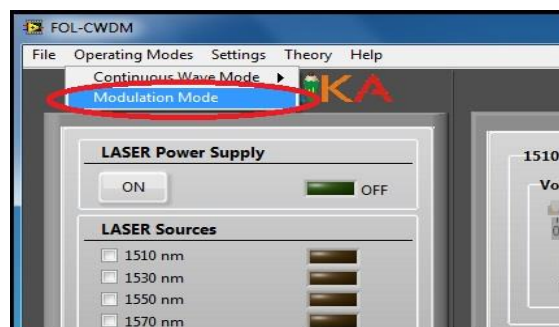
Function Generator.

PROCEDURE :

1. Setup the FOL-CWDM kit as explained in experiment 1.

To navigate to the modulation mode, go to;

Menu bar >> Operation modes >> Modulation

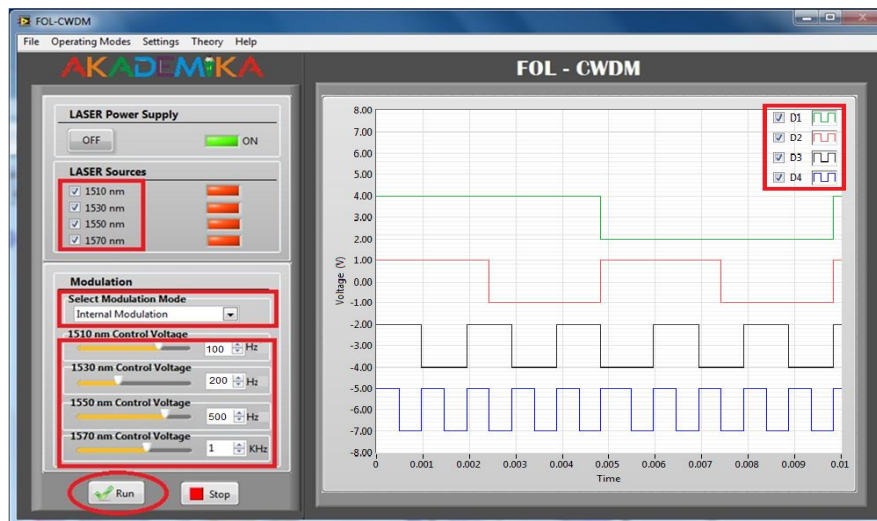


At the device end, connect the Detector O/P (electrical) externally to oscilloscope.

iii. Internal Modulation :

1. Select 'Internal Modulation' in Modulation mode selector dropdown.
2. To turn ON individual Laser diodes select the appropriate checkbox corresponding to laser wavelength.

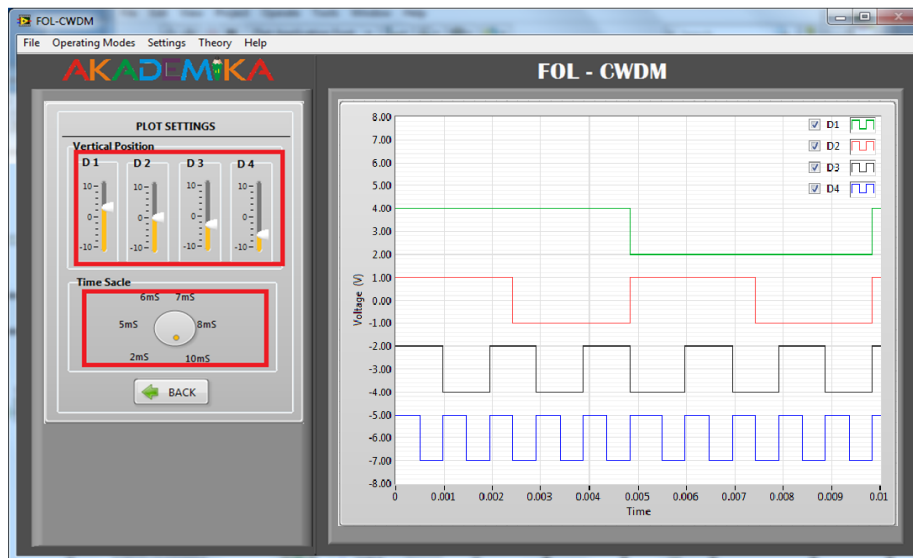
3. Set the control voltage sliders and modulation frequency among 100Hz, 200Hz, 500Hz and 1 KHz for each of the sources and click the 'RUN' Button.
4. Observe the modulated signals on the graph. These modulated signals can also be seen on the oscilloscope externally by connecting the electrical detector output to oscilloscope by using BNC to BNC cable.
5. Click 'STOP' button to stop the experiment.



6. To adjust the Time Scale and compare all of the four plots, go to;

Menu bar >> Settings >> Plot settings

7. Adjust Vertical position of the plots D1, D2, D3 D4 by using vertical sliders D1, D2, D3 and D4 respectively as shown in the figure.
8. Adjust the time base by using the Time scale knob.



iv. External Modulation :



1. Select 'External Modulation' in Modulation mode selector dropdown.
2. To turn ON individual Laser diodes select the appropriate checkbox corresponding to laser wavelength, adjust the control voltage sliders for each source and click the 'RUN' button.
3. Connect the BNC cables coming from the external Function Generator to the External I/Os on the back panel of the FOL-CWDM kit.

4. Connect BNC to BNC cable from electrical detectors D1, D2, D3 and D4 to the Oscilloscope.
5. The output signals will be seen on the Oscilloscope as shown below.
6. Click 'STOP' button to stop the modulation.

OBSERVATIONS :

Here, we studied and observed how to form communication link using digital modulation.

EXPERIMENT NO.5

EXPERIMENT NO : 5

NAME :

Multiplexing and De-multiplexing of optical wavelengths.

OBJECTIVE :

To multiplex and de-multiplex four optical wavelengths 1510nm, 1530nm, 1550nm and 1570nm using optical multiplexer and de-multiplexer with internal modulation.

APPARATUS :

FOL – CWDM Trainer Kit.

Power cord.

USB A to B cable.

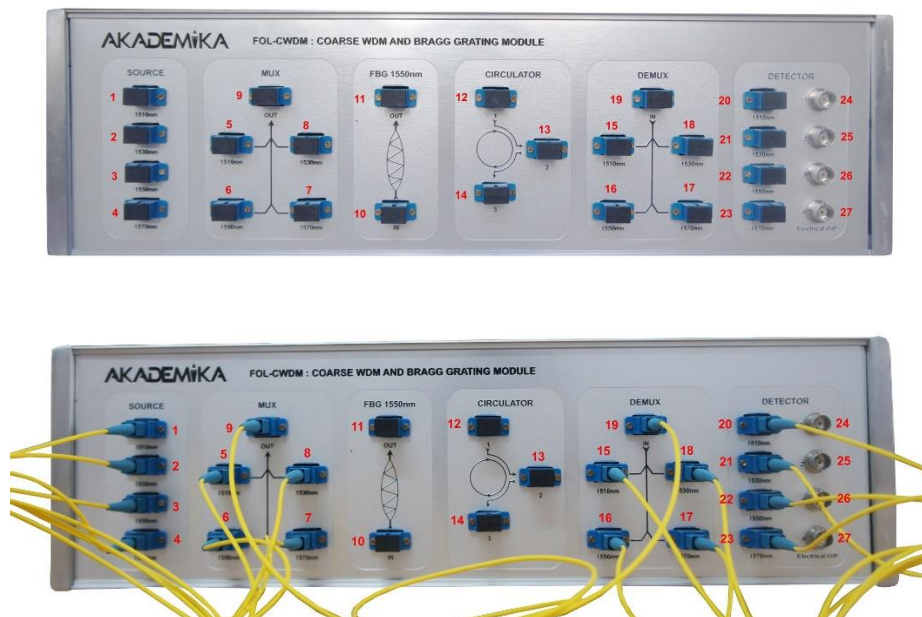
SC/PC – SC/PC fiber patch cords.

Fiber Optic Power Meter.

DSO.

PROCEDURE:

1. Setup the FOL-CWDM kit as explained in experiment 1 but with fiber connectivity as explained below.



2. Connect SC/PC – SC/PC fiber patch cords from source ports to MUX ports ; 1 → 5, 2 → 8, 3 → 6 and 4 → 7.
3. Connect MUX output (Port 9) to DEMUX input (Port 19).
4. Connect DEMUX ports to DETECTOR ports; 15 → 20, 18 → 21, 16 → 22 and 17 → 23. Remember detector sequence is not important. You can connect any source to any detector. It is user responsibility to keep the track of connections.
5. Connect electrical ports of DETECTOR to DSO.
6. Now run the system in internal modulation mode as explained in experiment 3.
7. If you connect MUX output (Port 9) to DETECTOR port and observe it's electrical output on DSO you will observe composite multiplexed signal from all four laser sources.
8. Shift the DEMUX optical ports to any of the DETECTOR optical ports and check for recovery of corresponding input wavelength.
9. Optical MUX and DE-MUX has fixed input wavelength ports. That means 1510nm port will carry only 1510nm wavelength and not any other.
10. You can swap wavelengths at the MUX or DE-MUX inputs to check the isolation of the channels.

OBSERVATION :

In this Experiment we studied Multiplexing and De-multiplexing of optical wavelengths.

EXPERIMENT NO.6

EXPERIMENT NO : 6

NAME :

Multiplexing , De-multiplexing and ADD – DROP of optical wavelengths using Bragg grating and optical circulator.

OBJECTIVE :

To demonstrate ADD-DROP technique of optical wavelengths using Bragg grating and optical circulator.

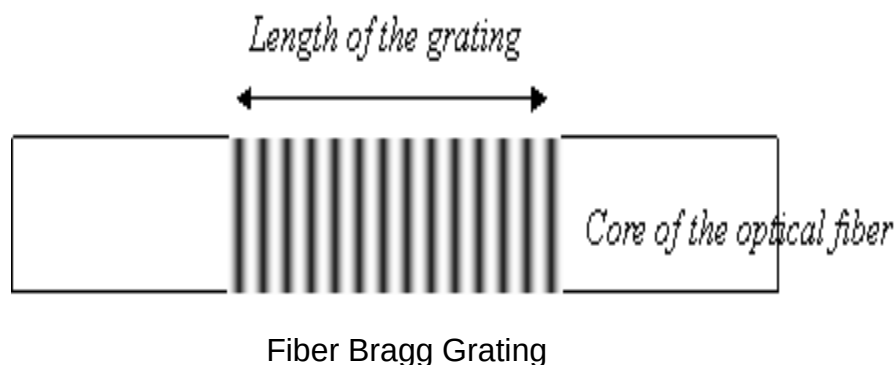
APPARATUS :

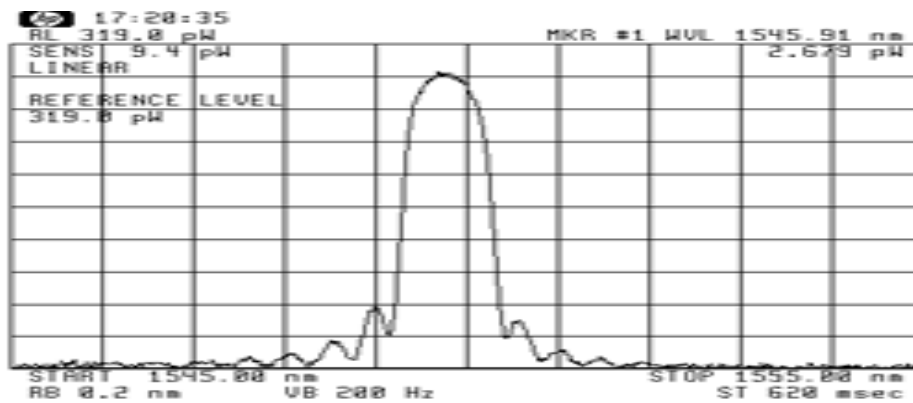
FOL – CWDM Trainer Kit.
Power cord.
USB A to B cable.
SC/PC – SC/PC fiber patch cords.
Optical Power Meter.
DSO.

THEORY:

Fiber Bragg Grating

Fiber Bragg Grating (FBG) is an optical fiber with a periodic variation of core refractive index along the fiber length. An FBG acts as a highly wavelength selective reflector, with a high reflectivity at a given central wavelength and the reflectivity dropping to very small values close to the central wavelength. The central wavelength, the peak value of reflectivity and the bandwidth of the reflection spectrum depends on the period of the refractive index modulation, on the index modulation of the grating and the length of the grating.





A Typical FBG Reflection Spectrum

If Λ is the period of index modulation, L the length of the grating and n the index modulation, then the central wavelength λ_c , peak reflectivity R and the bandwidth (the spectral width over which the reflectivity is high) are given approximately by

$$\lambda_c = 2n_e\Lambda$$

$$R = \tanh^2 \left(\frac{\pi \Delta n L}{\lambda_c} \right)$$

$$\Delta \lambda = \frac{\lambda_c^2}{n_e L} \left(1 + \left(\frac{\Delta n L}{\lambda_c} \right)^2 \right)$$

Here n_e is the effective index of the fundamental mode of the fiber.

As an example, let us consider an FBG with a period of 0.53 μm , an index modulation of 10^{-4} and a length of 5 mm made in a single mode fiber having a mode effective index of 1.45. Using the above set of equations we obtain the following characteristics of the grating: $\lambda_c = 1537$ nm, $R = 59\%$ and $\Delta \lambda = 0.36$ nm. By measuring λ_c , R and $\Delta \lambda$ of a grating, the values of Λ , L and n can be estimated from Equations.

FBGs find wide applications in optical fiber communication systems. They are used in laser transmitters for single frequency oscillation in optical add drop multiplexers, fiber lasers, and fiber optic sensors.

Optical circulator

An optical circulator is a multiport device with non reciprocal transmission characteristics. In Fig 4.3 we show a three port optical circulator. When light enters port 1 of the circulator, it exits from port 2. If light enters port 2 of the circulator, instead of its emerging from port 1 it now emerges from port 3 (showing non-reciprocity). Such a device finds wide applications in many areas such as dispersion compensation using FBGs, add/drop multiplexers etc. For example if an FBG with a central wavelength of λ_1 is placed at port 2 and if light at wavelengths λ_1 , λ_2 , and λ_3 , are incident on port 1 of the circulator, then out of the three wavelengths exiting from port 2, FBG reflects back wavelength λ_1 . This wavelength propagates back towards port 2 of the circulator and exits from port 3 while the wavelengths λ_2 , and λ_3 continue to propagate along port 2. Thus this acts as a drop filter for wavelength λ_1 .

The most important characteristics of a circulator are insertion loss and cross talk. These are defined as follows: If the power entering port 1 is P_1 and if the output at port 2 is P_2 , and that at port 3 is P_3 then the insertion loss is defined as

$$IL(dB) = -10 \log \left(\frac{P_2}{P_1} \right)$$

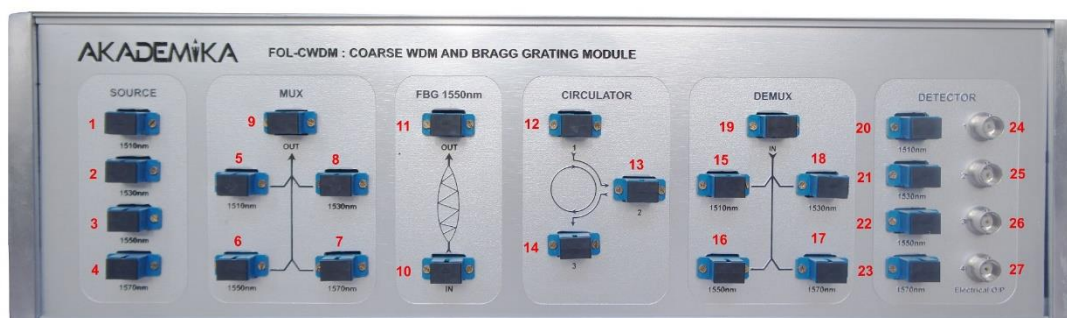
and the cross talk is defined as

$$CT(dB) = 10 \log \left(\frac{P_3}{P_2} \right)$$

Apart from these characteristics other important characteristics of a circulator include its polarization dependence in terms of polarization dependent loss (PDL) and polarization mode dispersion (PMD), the wavelength of operation and the power handling capacity. With additional equipment these characteristics can also be measured using the kit.

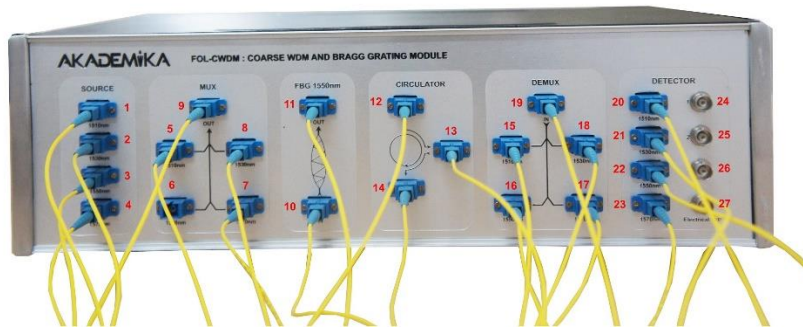
PROCEDURE :

Setup the FOL-CWDM kit as explained in experiment 1 but with fiber connectivity as explained below.



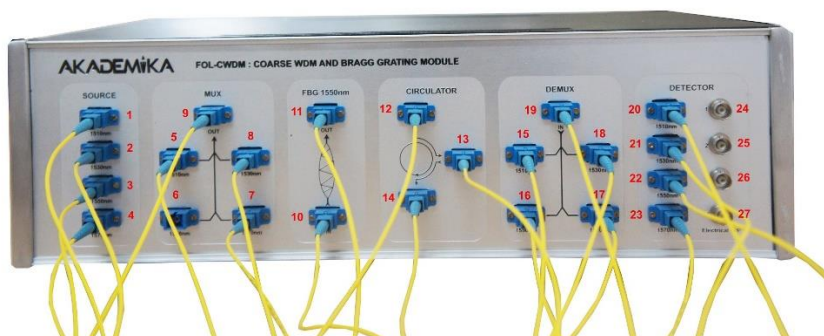
11. Connect SC/PC – SC/PC fiber patch cords from source ports to MUX ports ; 1 → 5, 2 → 8, 3 → 6 and 4 → 7.

1) To DROP wavelength 1550nm.



1. Connect MUX output (Port 9) to CIRCULATOR input (Port 12).
2. Connect FBG input (Port 10) to CIRCULATOR in / out (Port 13).
3. Connect FBG output (Port 11) to DE-MUX input (Port 19).
4. Bragg grating is designed to reflect 1550nm wavelength.
5. Circulator operates in forward direction i.e. energy input to port 12 appears at port 13 but not to port 14. Similarly energy input to port 13 appears at port 14 but not at port 12.
6. Here 4 wavelengths input to circulator port 12 are all passed to Bragg grating connected at port 13 of circulator.
7. Only 1550nm wavelength is reflected by Bragg grating and other are passed through towards DE-MUX input.
8. The reflected 1550nm wavelength acts as input to port 13 of the circulator which then comes out at port 14.
9. By connecting port 14 to detector input you can check the dropped wavelength.

2) To ADD wavelength 1550nm.



- 1.. Connect SC/PC – SC/PC fiber patch cords from source ports to MUX ports ; 1 → 5, 2 → 8 and 4 → 7. Hence we are multiplexing 1510nm, 1530nm and 1570nm wavelengths only.
2. Connect 1550nm source (Port 3) to Circulator input (Port 12).
3. Connect MUX output (Port 9) to Bragg grating input (Port 10).
4. Connect Bragg grating output (Port 11) to circulator (port 13).

5. Here 1550nm wavelength input to circulator port 12 is passed to Bragg grating and is reflected back to port 13 of the circulator.
6. This reflected wavelength acts as input to circulator at port 13 and comes out at port 14.
7. Also the three multiplexed wavelengths which are input to Bragg grating are directly passed to circulator port 13 where already reflected wavelength 1550nm is present.
8. Now all the four wavelengths at port 13 of the circulator come out at port 14.
9. Thus at port 14 of the circulator we get addition of multiplexed as well as reflected wavelength.
10. Finally connect port 14 of circulator to DE-MUX input (Port 19) and checked the de multiplexed output on the detectors.

RESULTS :

We observed Add-Drop of wavelength using Fiber Bragg Grating and Circulator.